Reinforcement Learning-Based Active Fault-Tolerant Control of Multi-Agent Systems

Sikai Shen, Xiaoyang Liu, and Wenwu Yu

***Abstract*—This article investigates the optimal tracking control problem for multi-agent systems (MASs) subject to actuator faults by synthesizing reinforcement learning (RL) with control barrier functions (CBFs), and active fault-tolerant control (AFTC). A novel integrated framework is proposed, wherein RL is utilized to achieve optimal tracking controller, while the AFTC scheme dynamically reconfigures the control law based on real-time fault diagnostic information. To guarantee operational safety, a CBF-based safety mechanism is embedded to maintain the forward invariance of a predefined safe set, ensuring the system operates within safe boundaries despite the presence of faults and external disturbances. In contrast to conventional quadratic programming (QP)-based approaches for enforcing safety constraints, the method offers a low-complexity solution, rendering it suitable for real-time implementation in large-scale MASs. Simulation results demonstrate the effectiveness of the proposed strategy in achieving stable consensus tracking and preserving system safety under fault conditions.**

***Impact Statement*—The increase of MASs is revolutionizing safety-critical domains like autonomous robotics and distributed control networks. In these applications, ensuring system integrity during unforeseen actuator failures is a paramount yet challenging goal. This article resolves this critical trade-off by introducing a unified framework that integrates RL for model-free optimal control, AFTC for dynamic fault compensation, and a novel low-complexity safety certificate based on CBFs. The key innovation is a method that provides rigorous, formal safety guarantees without the prohibitive computational overhead of conventional optimization-based approaches, making it possible to simultaneously achieve certified safety, optimality, and fault tolerance in a scalable manner.**

***Index Terms*—Active fault-tolerant control (AFTC), consensus, control barrier function (CBF), multi-agent systems (MASs), reinforcement learning (RL), safety.**

Received 9 January 2026; revised 5 March 2026; accepted 23 March 2026. Date of publication 30 March 2026; date of current version 31 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62276119, in part by the Natural Science Foundation of Jiangsu Province under Grant BK20241764, in part by the Postgraduate Research and Practice Innovation Program of Jiangsu Normal University under Grant 2024XKT2630, and in part by the Postgraduate Practice Innovation Program of Jiangsu Province under Grant SJCX25_1523. This article was recommended for publication by Associate Editor Hao Luo upon evaluation of the reviewers' comments. *(Corresponding author: Xiaoyang Liu.)*

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Digital Object Identifier 10.1109/TAI.2026.3678538

I. Introduction

In recent years, multi-agent systems (MASs) have gained significant attention, finding widespread application in both academic research and practical domains. A MAS is composed of multiple interconnected agents that coordinate their actions to accomplish collective objectives. This inherent flexibility and scalability make MASs suitable for diverse applications, including autonomous vehicles [1], robotics [2], and smart grids [3]. A primary challenge in the field of MASs is the design of distributed coordination protocols that ensure robust performance in dynamic and uncertain environments. Among the core research topics, consensus and formation control have been extensively studied [4], [5], [6], [7], [8], [9]. For instance, Li et al. [4] investigated leader–follower output consensus in heterogeneous MASs under a fully distributed control architecture. Concurrently, Gao et al. [5] addressed the secure formation control for heterogeneous MASs by integrating reinforcement learning (RL) and fuzzy logic to counteract communication attacks.

As the complexity of MASs increases, RL has emerged as a potent tool within control theory [10], [11], [12], [13], [14], offering a data-driven paradigm for solving optimization problems that are often intractable for traditional methods. The fundamental principle of RL involves an agent learning an optimal policy through iterative interactions with its environment, receiving feedback via reward or penalty signals. In contrast to conventional model-based techniques that necessitate precise system models, RL enables agents to learn directly from experience [15], which is particularly advantageous in scenarios where the system dynamics are unknown or difficult to model. For example, Liu et al. [15] proposed a data-driven RL method to address the optimal bipartite consensus problem of second-order MASs with unknown dynamics.

A prominent paradigm in RL is the actor-critic architecture [16], [17], where an "actor" is responsible for action selection (policy), and a "critic" evaluates these actions and provides feedback (value function). This feedback mechanism facilitates a continuous improvement cycle in the agent's decision-making process. The capacity of RL to manage system uncertainties and nonlinearities makes it an exceptionally suitable methodology for addressing control challenges in nonlinear MASs [18], [19]. For instance, Zhou et al. [18] applied RL to achieve adaptive optimal bipartite consensus control for nonlinear heterogeneous MASs via an event-triggered state feedback. In [19], RL was employed to design a distributed containment controller for nonlinear MASs under a dynamic event-triggered framework.

The occurrence of faults, arising from component failures or operation in harsh environments, is an inherent challenge in complex engineering systems. Historically, fault-tolerant control (FTC) strategies have predominantly relied on passive FTC (PFTC). PFTC schemes employ robust and adaptive control techniques to maintain system stability and performance in the presence of predefined faults [20], [21], [22], [23], [24], [25]. Du et al. [21] proposed a robust fault-tolerant H-infinity control framework for networked cascade control systems to address time delays and component failures. Similarly, an adaptive FTC approach based on predefined-time stability theory was proposed in [22] to handle actuator faults and mismatched disturbances. However, a significant limitation of PFTC is its static nature; while effective under anticipated fault conditions, it lacks the adaptability to respond dynamically to unforeseen fault characteristics.

In contrast, active FTC (AFTC) offers a more flexible and responsive solution. AFTC integrates fault detection and diagnosis (FDD) modules to identify and characterize faults in real time, subsequently reconfiguring the control strategy to accommodate the fault's effects [26], [27]. This proactive approach enables the system not only to tolerate faults but also to actively compensate for their impact, thereby preserving optimal performance. For example, Jia et al. [26] proposed an AFTC strategy, incorporating an adaptive error compensation mechanism to mitigate estimation errors from mismatched disturbances, ensuring asymptotic tracking in nonlinear systems. AFTC operates by adjusting the controller based on real-time fault information, a method particularly effective for the nonlinear systems prevalent in modern engineering [28], [29], [30]. These systems typically adhere to the separation principle, which allows for the independent design of fault detection, fault estimation (FE), and control modules, followed by their integration to achieve comprehensive fault tolerance [31], [32]. In [32], an AFTC scheme was proposed to handle intermittent faults in nonlinear systems, using fault diagnosis and adaptive compensation to maintain system stability.

In addition to fault tolerance, ensuring system safety has become a key concern in control system design, especially for systems operating in uncertain environments. Control barrier functions (CBFs) have emerged as an effective tool for enforcing safety [33], [34]. The core objective of the CBFs approach is to render a predefined safe set forward invariant, thereby guaranteeing that system trajectories, originating within this set, remain confined to it for all future time. This is accomplished by imposing a constraint, known as the control barrier condition (CBC), on the time derivative of the barrier function.

Initial research on CBFs focused on systems with a low relative degree, where the control input has a direct influence on the first derivative of the barrier function [35], [36]. For higher-order systems, where the control input affects the barrier function through higher-order derivatives, the application of standard CBFs techniques is nontrivial. To solve this, extended CBFs formulations have been developed [37], [38]. Furthermore, CBFs have been combined with backstepping-based

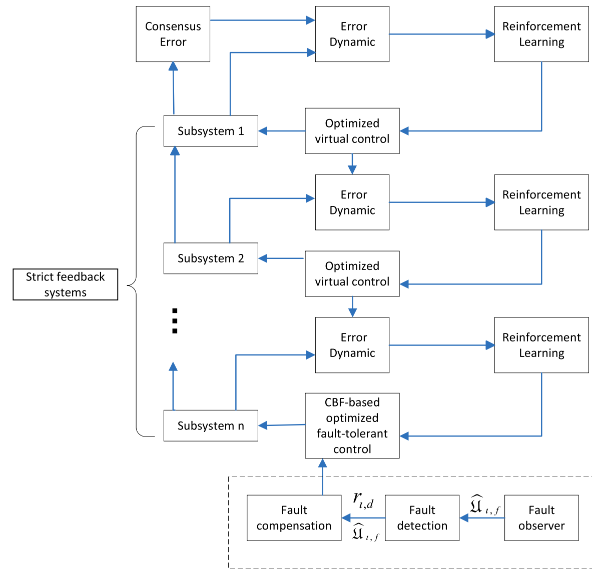


Fig. 1. Structure of the RL-based AFTC in ιth agent.

control methods to handle high-order system dynamics while keeping safety constraints in place [39], [40]. In [40], a low-complexity framework for safe control of higher-order MASs was proposed using CBFs based on funnel functions.

In practical implementations, CBF-based safety constraints are typically enforced through online optimization, most commonly through quadratic programming (QP), to satisfy the CBC at each time step [41], [42]. However, this approach can be computationally demanding, potentially posing feasibility challenges, particularly in the presence of modeling uncertainties or external disturbances. Additionally, the construction of suitable CBFs often requires accurate knowledge of the system dynamics, which can make the method sensitive to uncertainties and complicate the design process.

Despite the above progress, optimal and safe coordination for high-order nonlinear MASs under actuator faults remains insufficiently addressed. Existing RL-based schemes often improve performance but do not explicitly guarantee safety when faults enter the dynamics; meanwhile, QP-based CBC enforcement can impose notable computation and feasibility burdens in distributed settings. These gaps motivate a unified, low-complexity framework that integrates fault tolerance, safety guarantees, and online optimal control. The structure of the proposed strategy is shown in Fig. 1.

Motivated by the preceding discussions, this article addresses the problem of optimal and safe tracking control for MASs by integrating RL, CBFs, and AFTC. The primary contributions are threefold.

1) Compared with existing FTC strategies that rely on PFTC methods [21], [22], this article introduces an AFTC framework. Unlike PFTC, which lacks the adaptability to respond dynamically to faults, our AFTC scheme utilizes real-time FDD to adjust the control strategy dynamically, ensuring robust consensus tracking performance even during actuator failures.

2) Departing from conventional model-based control methods [26], [40], this article employs an RL-based actor-critic framework to solve the optimal tracking control problem for MASs. This allows the system to learn the optimal controller online, thereby enhancing overall performance.
3) A novel low-complexity, safety-guaranteed control method for MASs is proposed, founded on CBFs derived from funnel functions. In contrast to computationally demanding QP-based approaches [41], [42], our method circumvents online optimization, significantly reducing the computational burden while ensuring safety in the presence of high-order dynamics and disturbances.

The structure of this article is as follows. Section II formulates the preliminaries. Section III develops the AFTC scheme. Section IV presents the main results. Section V provides the simulations. Finally, Section VI concludes this article.

II. Preliminaries and Problem Formulations

A. Preliminaries on Graph Theory

In this article, an undirected connected graph $\mathbb{G} = \{\mathbb{V}, \mathbb{E}\}$ is utilized to represent the communication topology of the MASs. The adjacency matrix $\mathcal{B} = [b_{\iota\ell}] \in \mathbb{R}^{N\times N}$ is defined such that $b_{\iota\ell} > 0$ if an edge exists between node v_ι and node v_ℓ in the graph, i.e., $(v_\iota, v_\ell) \in \mathbb{E}$, and $b_{\iota\ell} = 0$ otherwise, with the diagonal elements $b_{\iota\iota} = 0$. Additionally, a diagonal matrix $\mathcal{P} = \mathrm{diag}\{p_1, p_2, \ldots, p_N\}$ is defined, where $p_\iota > 0$ if node ι can receive information from the leader $y_0(t)$, and $p_\iota = 0$ otherwise.

Assumption 1: The graph $\mathbb{G}$ is connected and contains a spanning tree with the leader node as its root.

Assumption 2: The leader's trajectory $y_0(t)$ and its derivative are bounded and continuous.

B. Preliminaries on Safety and CBF

In control theory, safety is commonly characterized by the forward invariance of a specified set in the state space. Consider the nonlinear system

$$\dot{\mathfrak{X}}(t) = \wp(\mathfrak{X}(t)) + \mathfrak{U}(t) + \varrho(t), \quad \mathfrak{X}(0) = \mathfrak{X}_0 \tag{1}$$

where $\mathfrak{X}(t)$ denotes the system state, $\mathfrak{U}(t)$ is the control input, and $\varrho(t)$ represents an external disturbance. If the function $\wp(\mathfrak{X}(t))$ is locally Lipschitz, then for any initial condition $\mathfrak{X}_0$, there exists a maximal time interval $[0, t_\tau)$ on which a unique solution $\mathfrak{X}(t)$ to (1) exists. Moreover, if system (1) is forward complete, then $t_\tau = +\infty$.

The safe set $\mathfrak{S} \subset \mathbb{R}^n$ is defined as a subset of the state space characterized by a superlevel set of a function, given by

$$\mathfrak{S} = \{\mathfrak{X} \in \mathbb{R}^n \mid \mathcal{E}(\mathfrak{X}) \geq 0\} \tag{2}$$

where $\mathcal{E}(\mathfrak{X})$ is a continuously differentiable function. The function $\mathcal{E}(\mathfrak{X})$ satisfies $(\partial\mathcal{E}/\partial\mathfrak{X})(\mathfrak{X}) \neq 0$, when $\mathcal{E}(\mathfrak{X}) = 0$.

Definition 1 [33]: A function $\mathcal{E}(\mathfrak{X})$ is a CBF for system (1) on a set $\mathfrak{S}$ as defined in (2) if there exists a function $\varpi(\cdot) \in \mathcal{K}_\infty^e$ such that for all $\mathfrak{X}$

$$\sup_{\mathfrak{U}} \left(\frac{\partial\mathcal{E}}{\partial\mathfrak{X}}\wp(\mathfrak{X}) + \frac{\partial\mathcal{E}}{\partial\mathfrak{X}}\mathfrak{U}(t) + \frac{\partial\mathcal{E}}{\partial\mathfrak{X}}\varrho(t) + \varpi\mathcal{E}(\mathfrak{X}) \right) \geq 0.$$

This inequality is referred to as the CBC.

Lemma 1 [33]: Consider a set $\mathfrak{S}$ in (2), and let $\mathcal{E}(\mathfrak{X})$ be a CBF for the system (1) on $\mathfrak{S}$ with the corresponding function $\varpi(t) \in \mathcal{K}_\infty^e$. Then, any Lipschitz continuous controller $\mathfrak{U}$ that satisfies $\mathfrak{U} \in \Phi_{\mathfrak{U}}(\mathfrak{X})$, where $\Phi_{\mathfrak{U}}(\mathfrak{X}) = \{\mathfrak{U} \mid (\partial\mathcal{E}/(\partial\mathfrak{X})\wp(\mathfrak{X}) + (\partial\mathcal{E}/\partial\mathfrak{X})\mathfrak{U}(t) + (\partial\mathcal{E}/\partial\mathfrak{X})\varrho(t) + \varpi(t)\mathcal{E}(\mathfrak{X})) \geq 0\}$, will ensure that the set $\mathfrak{S}$ remains forward invariant.

Consider the nonlinear system

$$\dot{\aleph}(t) = \Theta(t, \aleph(t)), \quad \aleph(0) = \aleph^0 \in \Lambda_\aleph \tag{3}$$

where the Θ is locally Lipschitz with respect to $\aleph(t)$, and is continuous and locally integrable with respect to time for each fixed $\aleph(t) \in \Lambda_\aleph$. The set $\Lambda_\aleph \subset \mathbb{R}^n$ is assumed to be nonempty and open.

Definition 2 [43]: A solution $\aleph(t)$ of system (3) is called maximal if it cannot be extended to the right while still satisfying system (3).

Lemma 2 [43]: Under the conditions in (3), for any initial condition $\aleph^0 \in \Lambda_\aleph$, there exists a unique maximal solution $\aleph(t)$ defined on an interval $[0, t_\tau)$, where $t_\tau \in \mathbb{R}_{\geq 0} \cup \{+\infty\}$. Moreover, this solution satisfies $\aleph(t) \in \Lambda_\aleph$ for all $t \in [0, t_\tau)$.

Lemma 3 [43]: For a maximal solution $\aleph(t) : [0, t_\nu) \to \Lambda_\aleph$ of (3) defined on a finite interval with $t_\nu < \infty$, and for any compact set $\Lambda' \subset \Lambda_\aleph$, there exists a time instant $t_j \in [0, t_\nu)$ such that $\aleph(t_j) \notin \Lambda'$.

C. Problem Formulation

Consider a MAS where each agent is described by the following nth order nonlinear dynamics with an actuator fault

$$\begin{cases} \dot{\mathfrak{X}}_{\iota,\ell}(t) = \mathfrak{X}_{\iota,\ell+1}(t) + \wp_{\iota,\ell}(\bar{\mathfrak{X}}_{\iota,\ell}(t)) + \varrho_{\iota,\ell}(t) \\ \dot{\mathfrak{X}}_{\iota,n}(t) = b_\iota(\mathfrak{U}_{\iota,0}(t) + \mathfrak{U}_{\iota,f}(t)) + \wp_{\iota,n}(\bar{\mathfrak{X}}_{\iota,n}(t)) + \varrho_{\iota,n}(t) \\ y_\iota(t) = \mathfrak{X}_{\iota,1}(t) \end{cases} \tag{4}$$

where $\ell = 1, \ldots, n-1$, $\bar{\mathfrak{X}}_{\iota,\ell}(t) = [\mathfrak{X}_{\iota,1}(t), \ldots, \mathfrak{X}_{\iota,\ell}(t)]^T \in \mathbb{R}^\ell$ denotes the state vector. The terms $y_\iota(t) \in \mathbb{R}$ and $\mathfrak{U}_{\iota,0}(t) \in \mathbb{R}$ represent the system output and the nominal control input, respectively. The term $\varrho_{\iota,\ell}(t) \in \mathbb{R}$ represents an external disturbance, and $\wp_{\iota,\ell}(\bar{\mathfrak{X}}_{\iota,\ell}(t)) \in \mathbb{R}$ is a known nonlinear function. The constant b_ι denotes the input gain, and $\mathfrak{U}_{\iota,f}(t)$ denotes a time-varying actuator fault.

Assumption 3: The actuator fault $\mathfrak{U}_{\iota,f}(t)$ is bounded, and its rate of change satisfies $0 < |\dot{\mathfrak{U}}_{\iota,f}(t)| < \bar{\mathfrak{U}}_{\iota,f}$, where $\bar{\mathfrak{U}}_{\iota,f}$ is a constant. The disturbance $\varrho_{\iota,\ell}(t)$ is bounded such that $|\varrho_{\iota,\ell}(t)| < \hbar_{\iota,\ell}$ for some positive constant $\hbar_{\iota,\ell}$.

Assumption 4: The nonlinear function $\wp_{\iota,\ell}$ is known and satisfies the Lipschitz condition

$$|\wp_{\iota,\ell}(\bar{\mathfrak{X}}_{\iota,1}) - \wp_{\iota,\ell}(\bar{\mathfrak{X}}_{\iota,2})| \leq L_{Z,\iota}|\bar{\mathfrak{X}}_{\iota,1} - \bar{\mathfrak{X}}_{\iota,2}|$$

where $L_{Z,\iota}$ is a positive constant.

III. Fault Detection and Estimation

To implement the AFTC strategy, real-time fault detection and estimation are crucial for maintaining system stability and performance. This section details the design of the fault detection and estimation modules.

A. Fault Detection

The fault detection observer is designed as follows:

$$\begin{cases}\dot{\hat{\mathfrak{X}}}_{\iota,nd}(t)=\wp_{\iota,n}(\bar{\mathfrak{X}}_{\iota,n-1},\hat{\mathfrak{X}}_{\iota,nd})+b_\iota\mathfrak{U}_{\iota,0}(t)+\lambda_\iota\tilde{\mathfrak{X}}_{\iota,nd}(t)\\ \tilde{\mathfrak{X}}_{\iota,nd}(t)=\mathfrak{X}_{\iota,n}(t)-\hat{\mathfrak{X}}_{\iota,nd}(t)\\ r_{\iota,d}(t)=|\tilde{\mathfrak{X}}_{\iota,nd}(t)|\end{cases} \tag{5}$$

where $\hat{\mathfrak{X}}_{\iota,nd}(t)$ is the state of the detection observer, initialized at $\hat{\mathfrak{X}}_{\iota,nd}(0)=0$. The design parameter λ_ι is a positive constant chosen to satisfy $\lambda_\iota>L_{Z,\iota}$. The residual signal $r_{\iota,d}(t)$ is constructed from the estimation error $\tilde{\mathfrak{X}}_{\iota,nd}(t)$ and serves as the basis for fault identification.

The fault detection logic is formulated as follows:

$$\begin{cases}r_{\iota,d}(t)>R_{th,\iota}, \text{fault detected}\\ r_{\iota,d}(t)\le R_{th,\iota}, \text{no fault detected}\end{cases} \tag{6}$$

where the detection threshold $R_{th,\iota}$ is set as $R_{th,\iota}=\sup_{\mathfrak{U}_{\iota,f}=0} r_{\iota,d}(t)$.

Theorem 1: For system (4) with the fault–detection observer (5) and rule (6), a fault is guaranteed to be detected if the following condition holds:

$$\mathfrak{U}_{\iota,f}(t)>\frac{2\lambda_\iota\hbar_{\iota,n}}{\lambda_\iota-L_{Z,\iota}}. \tag{7}$$

Proof: Define $\hat{\wp}_{\iota,nd}=\wp_{\iota,n}(\bar{\mathfrak{X}}_{\iota,n-1},\hat{\mathfrak{X}}_{\iota,nd})$ and introduce the positive variable

$$L_{\wp,\iota}(t)=\frac{|\wp_{\iota,n}(\bar{\mathfrak{X}}_{\iota,n})-\hat{\wp}_{\iota,nd}|}{|\tilde{\mathfrak{X}}_{\iota,nd}(t)|}. \tag{8}$$

Since the inequality $|\wp_{\iota,n}(\bar{\mathfrak{X}}_{\iota,n})-\hat{\wp}_{\iota,nd}|\le L_{Z,\iota}|\tilde{\mathfrak{X}}_{\iota,nd}(t)|$ holds, we obtain $L_{\wp,\iota}(t)\le L_{Z,\iota}$.

Taking time derivative of $\tilde{\mathfrak{X}}_{\iota,nd}(t)$ yields

$$\begin{aligned}\dot{\tilde{\mathfrak{X}}}_{\iota,nd}(t)=&\wp_{\iota,n}-\hat{\wp}_{\iota,nd}-\lambda_\iota\tilde{\mathfrak{X}}_{\iota,nd}(t)\\&+b_\iota\mathfrak{U}_{\iota,f}(t)+\varrho_{\iota,n}(t).\end{aligned} \tag{9}$$

Substituting (8) into (9) yields

$$\dot{\tilde{\mathfrak{X}}}_{\iota,nd}(t)=-(\lambda_\iota\pm L_{\wp,\iota}(t))\tilde{\mathfrak{X}}_{\iota,nd}(t)+b_\iota\mathfrak{U}_{\iota,f}(t)+\varrho_{\iota,n}(t). \tag{10}$$

Before the fault occurs, $\tilde{\mathfrak{X}}_{\iota,nd}(0)=0$ and $\mathfrak{U}_{\iota,f}=0$. From (10) one has

$$|\tilde{\mathfrak{X}}_{\iota,nd}(t)|\le\frac{|\varrho_{\iota,n}(t)|}{\lambda_\iota-L_{Z,\iota}}\le\frac{\hbar_{\iota,n}}{\lambda_\iota-L_{Z,\iota}}, t\in[0,t_f).$$

When $t>t_f$, one can get

$$\begin{aligned}&\frac{|\mathfrak{U}_{\iota,f}(t)-\hbar_{\iota,n}|}{\lambda_\iota+L_{Z,\iota}}-\frac{\mathfrak{U}_{\iota,f}(t)+2\hbar_{\iota,n}}{\lambda_\iota-L_{Z,\iota}}e^{-(\lambda_\iota-L_{Z,\iota})(t-t_f)}\\&\le|\tilde{\mathfrak{X}}_{\iota,nd}(t)|\le\frac{\hbar_{\iota,n}+\mathfrak{U}_{\iota,f}(t)}{\lambda_\iota-L_{Z,\iota}}\end{aligned}$$

and

$$|\tilde{\mathfrak{X}}_{\iota,nd}(t)|\ge\frac{|\mathfrak{U}_{\iota,f}(t)-\hbar_{\iota,n}|}{\lambda_\iota+L_{Z,\iota}}, t\to\infty.$$

If (7) is satisfied, one can obtain

$$\frac{\hbar_{\iota,n}}{\lambda_\iota-L_{Z,\iota}}<\frac{|\mathfrak{U}_{\iota,f}(t)-\hbar_{\iota,n}|}{\lambda_\iota+L_{Z,\iota}}$$

which means if $r_{\iota,d}(t)>\hbar_{\iota,n}/(\lambda_\iota-L_{Z,\iota})$, the fault occurs. ■

B. FE

Once a fault is detected, a fault estimator is activated to quantify its magnitude. The estimator is constructed as follows:

$$\begin{cases}\dot{\hat{\mathfrak{X}}}_{\iota,ne}(t)=\wp_{\iota,n}(\bar{\mathfrak{X}}_{\iota,n-1},\hat{\mathfrak{X}}_{\iota,ne})\\ \qquad\qquad +b_\iota\mathfrak{U}_{\iota,0}(t)+\hat{\mathfrak{U}}_{\iota,f}(t)+L_\iota(\mathfrak{X}_{\iota,n}(t)-\hat{\mathfrak{X}}_{\iota,ne}(t))\\ \dot{\hat{F}}_\iota(t)=-G_\iota(\wp_{\iota,n}(\bar{\mathfrak{X}}_{\iota,n-1},\hat{\mathfrak{X}}_{\iota,ne})\\ \qquad\qquad +b_\iota\mathfrak{U}_{\iota,0}(t)+G_\iota\hat{\mathfrak{X}}_{\iota,ne}(t))-G_\iota\hat{F}_\iota(t)\\ F_\iota(t)=\mathfrak{U}_{\iota,f}(t)-G_\iota\mathfrak{X}_{\iota,n}(t)\\ \hat{\mathfrak{U}}_{\iota,f}(t)=\hat{F}_\iota(t)+G_\iota\hat{\mathfrak{X}}_{\iota,ne}(t)\end{cases} \tag{11}$$

where $\hat{\mathfrak{X}}_{\iota,ne}(t)$ is an auxiliary state, while $\hat{F}_\iota(t)$ and $\hat{\mathfrak{U}}_{\iota,f}(t)$ are the estimates of $F_\iota(t)$ and $\mathfrak{U}_{\iota,f}(t)$, respectively. The parameters $G_\iota>0$ and L_ι are design constants.

Theorem 2: Consider system (4) equipped with the estimator in (11). If $\sigma_\iota>0$ and $G_\iota>0$, then the estimation error $\tilde{\mathfrak{U}}_{\iota,f}$ converges exponentially to a bounded invariant set containing the origin, provided there exist constants $\rho_{\iota,1}>0$, $\rho_{\iota,2}>0$ and a gain L_ι such that

$$P_\iota=\begin{bmatrix}\frac{1}{\rho_{\iota,1}}\left(L_\iota-G_\iota-L_{Z,\iota}-\frac{\sigma_\iota}{2}\right)-\frac{\sigma_\iota\rho_{\iota,2}L_{Z,\iota}^2}{2} & *\\ \frac{1}{2}\left(\rho_{\iota,2}G_\iota^2-\frac{1}{\rho_{\iota,1}}\right) & \rho_{\iota,2}\left(G_\iota-\frac{1}{\sigma_\iota}G_\iota^2-\frac{1}{2\sigma_\iota}\right)\end{bmatrix}>0. \tag{12}$$

Proof: Consider the Lyapunov function as

$$V_{e,\iota}(t)=\frac{1}{2\rho_{\iota,1}}\tilde{\mathfrak{X}}_{\iota,ne}^2(t)+\frac{\rho_{\iota,2}}{2}\tilde{F}_\iota^2(t) \tag{13}$$

where $\tilde{\mathfrak{X}}_{\iota,ne}(t)=\mathfrak{X}_{\iota,n}(t)-\hat{\mathfrak{X}}_{\iota,ne}(t)$. Taking the time derivative of $V_{e,\iota}(t)$ yields

$$\begin{aligned}\dot{V}_{e,\iota}(t)=&\frac{1}{\rho_{\iota,1}}\tilde{\mathfrak{X}}_{\iota,ne}(t)\big(\wp_{\iota,n}(\bar{\mathfrak{X}}_{\iota,n})-\wp_{\iota,n}(\bar{\mathfrak{X}}_{\iota,n-1},\hat{\mathfrak{X}}_{\iota,ne})\\&+\varrho_{\iota,n}(t)-L_\iota\tilde{\mathfrak{X}}_{\iota,ne}(t)+\tilde{F}_\iota(t)+G_\iota\tilde{\mathfrak{X}}_{\iota,ne}(t)\big)\\&+\rho_{\iota,2}\tilde{F}_\iota(t)\big(\dot{\mathfrak{U}}_{\iota,f}(t)+G_\iota(\wp_{\iota,n}(\bar{\mathfrak{X}}_{\iota,n-1},\hat{\mathfrak{X}}_{\iota,ne})\\&-\wp_{\iota,n}(\bar{\mathfrak{X}}_{\iota,n}))-G_\iota\varrho_{\iota,n}(t)-G_\iota\tilde{F}_\iota(t)+G_\iota^2\tilde{\mathfrak{X}}_{\iota,ne}(t)\big).\end{aligned} \tag{14}$$

According to Assumptions (3) and (4) and Young's inequality, it has

$$|\wp_{\iota,n}(\bar{\mathfrak{X}}_{\iota,n})-\wp_{\iota,n}(\bar{\mathfrak{X}}_{\iota,n-1},\hat{\mathfrak{X}}_{\iota,ne})|\le L_{Z,\iota}|\tilde{\mathfrak{X}}_{\iota,ne}(t)| \tag{15}$$

$$\tilde{\mathfrak{X}}_{\iota,ne}(t)\varrho_{\iota,n}(t)\le\frac{\sigma_\iota}{2}\tilde{\mathfrak{X}}_{\iota,ne}^2(t)+\frac{1}{2\sigma_\iota}\hbar_{\iota,n}^2 \tag{16}$$

$$\tilde{F}_\iota(t)\dot{\mathfrak{U}}_{\iota,f}(t)\le\frac{1}{2\sigma_\iota}\tilde{F}_\iota^2(t)+\frac{\sigma_\iota}{2}b_\iota^2\bar{\mathfrak{U}}_{\iota,f}^2 \tag{17}$$

$$-G_\iota\varrho_{\iota,n}(t)\tilde{F}_\iota(t)\le\frac{1}{2\sigma_\iota}(G_\iota\tilde{F}_\iota(t))^2+\frac{\sigma_\iota}{2}\hbar_{\iota,n}^2 \tag{18}$$

$$\begin{aligned}&\tilde{F}_\iota(t)G_\iota(\wp_{\iota,n}(\bar{\mathfrak{X}}_{\iota,n-1},\hat{\mathfrak{X}}_{\iota,ne})-\wp_{\iota,n}(\bar{\mathfrak{X}}_{\iota,n}))\\&\le\frac{1}{2\sigma_\iota}(\tilde{F}_\iota(t)G_\iota)^2+\frac{\sigma_\iota}{2}L_{Z,\iota}^2\tilde{\mathfrak{X}}_{\iota,ne}^2(t).\end{aligned} \tag{19}$$

Substituting (15)-(19) into (14) yields

$$\dot{V}_{e,\iota}(t)\le-[\tilde{\mathfrak{X}}_{\iota,ne}(t),\tilde{F}_\iota(t)]P_\iota[\tilde{\mathfrak{X}}_{\iota,ne}(t),\tilde{F}_\iota(t)]^T+\Upsilon_\iota \tag{20}$$

where $\Upsilon_\iota=(1/2\sigma_\iota)\hbar_{\iota,n}^2+(\sigma_\iota/2)b_\iota^2\bar{\mathfrak{U}}_{\iota,f}^2+(\sigma_\iota/2)\hbar_{\iota,n}^2$.

From (12) and (20), it follows that

$$\begin{aligned}\dot{V}_{e,\iota}(t)&\le-\lambda_\iota^{\min}[P_\iota]\tilde{\mathfrak{X}}_{\iota,ne}^2(t)-\lambda_\iota^{\min}[P_\iota]\tilde{F}_\iota^2(t)+\Upsilon_\iota,\\&\le-\kappa_\iota V_{e,\iota}(t)+\Upsilon_\iota\end{aligned} \tag{21}$$

where $\kappa_\iota=2\lambda_\iota^{\min}[P_\iota]/\max\{1/\rho_{\iota,1},\rho_{\iota,2}\}$.

Combine the Lyapunov function candidates as follows:

$$V_e(t)=\sum_{\iota=1}^{N}V_{e,\iota}(t) \tag{22}$$

let $\kappa=\min\{2\lambda_1^{\min}[P_1]/\max\{1/\rho_{1,1},\rho_{1,2}\},\ldots,2\lambda_N^{\min}[P_N]/\max\{1/\rho_{N,1},\rho_{N,2}\}\}$, $\Upsilon=\sum_{\iota=1}^{N}(1/2\sigma_\iota\hbar_{\iota,n}^2+(\sigma_\iota/2)b_\iota^2\bar{\mathfrak{U}}_{\iota,f}^2+(\sigma_\iota/2)\hbar_{\iota,n}^2)$. The time derivative of (22) is given by

$$\dot{V}_e(t)\le-\kappa V_e(t)+\Upsilon.$$

■

Remark 1: Compared to PFTC schemes [21], [22], the proposed AFTC improves system efficiency by activating fault compensation mechanism only upon fault detection. PFTC applies compensation continuously, leading to inefficient allocation of control effort in fault-free conditions. By contrast, AFTC conserves control resources and improves performance during fault events by applying compensation only when necessary.

IV. Main Results

This section details the design of the controller and presents the stability and safety analysis of MASs using RL. It establishes conditions for achieving leader-following control, ensuring system stability, and guaranteeing safety through CBFs, even in the presence of faults and disturbances.

A. Stability Analysis

The consensus tracking error is defined in a step-by-step manner

$$\begin{aligned}&\zeta_{\iota,1}(t)=\sum_{l\in N_\iota}b_{\iota l}(y_\iota(t)-y_l(t))+p_\iota(y_\iota(t)-y_0(t))\\&\zeta_{\iota,\ell}(t)=\mathfrak{X}_{\iota,\ell}(t)-\hat{\alpha}_{\iota,\ell-1}^*(t),\ell=2,\ldots,n\end{aligned} \tag{23}$$

where $\hat{\alpha}_{\iota,\ell-1}^*(t)$ is the optimized virtual control signal derived at step $\ell-1$, and $y_0(t)$ is the leader's output trajectory.

To enforce prescribed performance bounds, we introduce the following coordinate transformation:

$$\aleph_{\iota,\ell}(t)=\frac{\zeta_{\iota,\ell}(t)}{\Psi_{\iota,\ell}(t)} \tag{24}$$

where $\Psi_{\iota,\ell}(t)$ is a funnel function defined as $\Psi_{\iota,\ell}(t):\mathbb{R}_{\ge0}\mapsto[\underline{\Psi}_{\iota,\ell},\overline{\Psi}_{\iota,\ell}]$ with $\overline{\Psi}_{\iota,\ell}>\underline{\Psi}_{\iota,\ell}>0$, and it satisfies the initial feasibility condition $\Psi_{\iota,\ell}(0)>|\zeta_{\iota,\ell}(0)|$. The funnel function $\Psi_{\iota,\ell}(t)$ is assumed to be continuous and differentiable, and $\dot{\Psi}_{\iota,\ell}(t)$ is bounded. If $\zeta_{\iota,\ell}(t)/\Psi_{\iota,\ell}(t)<1$ for all t, then the consensus error $\zeta_{\iota,\ell}(t)$ is said to satisfy the funnel performance [44].

Theorem 3: For the MASs (4) subject to actuator faults, under the proposed distributed controller synthesis involving the actor-critic structure and the AFTC mechanism (32), (41), (52), all signals in the closed-loop signals are uniformly ultimately bounded. Moreover, the leader-following consensus error satisfies the prescribed funnel performance for all time.

Let $\aleph=[\aleph_1^T,\ldots,\aleph_N^T]^T$, where $\aleph_\iota=[\aleph_{\iota,1},\ldots,\aleph_{\iota,n}]^T$. The stability of the closed-loop system is analyzed by invoking Lemma (2) and Lemma (3) for system (3). For this analysis, the following open set is introduced:

$$\Lambda_\aleph=\underbrace{(-1,1)\times\cdots\times(-1,1)}_{N\times n}.$$

The proof establishes existence and uniqueness of a maximal solution $\aleph(t)$ satisfying $\aleph(t)\in\Lambda_\aleph$ for $t\in[0,t_\tau)$, proves boundedness of all closed-loop signals with $\aleph(t)\in\Lambda'\subset\Lambda_\aleph$, and finally extends the solution to $t_\tau=+\infty$ by contradiction.

Proof: Since $\Psi_{\iota,\ell}(0)>|\zeta_{\iota,\ell}(0)|$, it follows that $\aleph(0)\in\Lambda_\aleph$, and Lemma (2) guarantees the existence of a unique maximal solution $\aleph(t)$ that remains in $\Lambda_\aleph$ for $t\in[0,t_\tau)$. Consequently, $\aleph_{\iota,k}(t)=(\zeta_{\iota,k}(t)/\Psi_{\iota,k}(t))\in(-1,1),\forall t\in[0,t_\tau)$.

Step 1: Taking the derivative of $\aleph_{\iota,1}(t)$ yields

$$\begin{aligned}\dot{\aleph}_{\iota,1}(t)=\Psi_{\iota,1}^{-1}[&(b_\iota+p_\iota)(\mathfrak{X}_{\iota,2}(t)+\wp_{\iota,1}(t)+\varrho_{\iota,1}(t))-p_\iota\dot{y}_0(t)\\&-\sum_{l\in N_\iota}b_{\iota l}(\mathfrak{X}_{l,2}(t)+\wp_{l,1}(t)+\varrho_{l,1}(t))-\dot{\Psi}_{\iota,1}(t)\aleph_{\iota,1}(t)].\end{aligned} \tag{25}$$

Denote the performance index function for the above subsystem as follows:

$$\mathcal{D}_{\iota,1}(\aleph_{\iota,1})=\int_t^\infty c_{\iota,1}(\aleph_{\iota,1}(s),\alpha_{\iota,1}(s))ds$$

where $c_{\iota,1}(\aleph_{\iota,1},\alpha_{\iota,1})=\aleph_{\iota,1}^2+\alpha_{\iota,1}^2$ is the cost function, $\alpha_{\iota,1}$ is the virtual control.

Then, the optimal performance index is

$$\mathcal{D}_{\iota,1}^*(\aleph_{\iota,1})=\int_t^\infty c_{\iota,1}(\aleph_{\iota,1}(s),\alpha_{\iota,1}^*(s))ds. \tag{26}$$

By treating $\mathfrak{X}_{\iota,2}(t)$ as the optimal virtual control $\alpha_{\iota,1}^*$, the Hamilton–Jacobi–Bellman (HJB) equation is obtained by taking the time derivative along (25) on both sides of the (26)

$$\begin{aligned}&H_{\iota,1}(\aleph_{\iota,1},\alpha_{\iota,1}^*,\mathcal{D}_{\iota,1}^*)\\&=c_{\iota,1}(\aleph_{\iota,1},\alpha_{\iota,1}^*)+\frac{d\mathcal{D}_{\iota,1}^*(\aleph_{\iota,1})}{d\aleph_{\iota,1}}\dot{\aleph}_{\iota,1}(t)\\&=\aleph_{\iota,1}^2(t)+\alpha_{\iota,1}^{*2}+\frac{d\mathcal{D}_{\iota,1}^*(\aleph_{\iota,1})}{d\aleph_{\iota,1}}\Psi_{\iota,1}^{-1}[(b_\iota+p_\iota)(\alpha_{\iota,1}^*(t)\\&+\wp_{\iota,1}(t)+\varrho_{\iota,1}(t))-p_\iota\dot{y}_0(t)-\sum_{l\in N_\iota}b_{\iota l}(\mathfrak{X}_{l,2}(t)\\&+\wp_{l,1}(t)+\varrho_{l,1}(t))-\dot{\Psi}_{\iota,1}(t)\aleph_{\iota,1}(t)]=0.\end{aligned}$$

By solving $\partial H_{\iota,1}/\partial \alpha^*_{\iota,1} = 0$, one has

$$\alpha^*_{\iota,1} = -\frac{1}{2}\Psi^{-1}_{\iota,1}(b_\iota + p_\iota)\frac{d\mathcal{D}^*_{\iota,1}(\aleph_{\iota,1})}{d\aleph_{\iota,1}}.$$

To achieve leader-following tracking control, $d\mathcal{D}^*_{\iota,1}(\aleph_{\iota,1})/d\aleph_{\iota,1}$ is reformulated as follows:

$$\frac{d\mathcal{D}^*_{\iota,1}(\aleph_{\iota,1})}{d\aleph_{\iota,1}} = \frac{2\lambda_{\iota,1}\aleph_{\iota,1}(t)}{\Psi^{-1}_{\iota,1}(b_\iota + p_\iota)}\frac{1}{1-\aleph^2_{\iota,1}(t)} + \frac{2\wp_{\iota,1}(t)}{\Psi^{-1}_{\iota,1}(b_\iota + p_\iota)} + \frac{-2}{\Psi^{-1}_{\iota,1}(b_\iota + p_\iota)^2}\left(\sum_{l\in N_\iota} b_{\iota l}(\mathfrak{X}_{l,2}(t) + \wp_{l,1}(t)) + p_\iota \dot{y}_0(t)\right) + \frac{1-\aleph^2_{\iota,1}(t)}{\Psi^{-2}_{\iota,1}(b_\iota + p_\iota)^2}\mathcal{D}^o_{\iota,1} \tag{27}$$

where $\lambda_{\iota,1} > 0$ is a constant, and $\mathcal{D}^o_{\iota,1} = (\Psi^{-1}_{\iota,1}(b_\iota + p_\iota)/1 - \aleph^2_{\iota,1}(t))(-2\lambda_{\iota,1}\aleph_{\iota,1}(t) - 2\wp_{\iota,1}(t)) + (2/(1-\aleph^2_{\iota,1}(t))\Psi_{\iota,1})(\sum_{l\in N_\iota} b_{\iota l}(\mathfrak{X}_{l,2}(t) + \wp_{l,1}(t)) + p_\iota \dot{y}_0(t)) + (\Psi^{-2}_{\iota,1}(b_\iota + p_\iota)^2/1 - \aleph^2_{\iota,1}(t))(d\mathcal{D}^*_{\iota,1}(\aleph_{\iota,1})/d\aleph_{\iota,1})$.

Therefore, the optimal virtual controller $\alpha^*_{\iota,1}(t)$ can be reformulated as follows:

$$\alpha^*_{\iota,1}(t) = -\lambda_{\iota,1}\frac{\aleph_{\iota,1}(t)}{1-\aleph^2_{\iota,1}(t)} - \wp_{\iota,1}(t) + \frac{1}{b_\iota + p_\iota} \times \left(\sum_{l\in N_\iota} b_{\iota l}(\mathfrak{X}_{l,2}(t) + \wp_{l,1}(t)) + p_\iota \dot{y}_0(t)\right) - \frac{1-\aleph^2_{\iota,1}(t)}{2\Psi^{-1}_{\iota,1}(b_\iota + p_\iota)}\mathcal{D}^o_{\iota,1}. \tag{28}$$

However, the virtual control (28) includes an unknown continuous term $\mathcal{D}^o_{\iota,1}$, which is approximated by a neural network over a compact set as follows:

$$\mathcal{D}^o_{\iota,1} = \mathcal{Q}^{*T}_{\iota,1}\mathfrak{W}_{\iota,1}(\aleph_{\iota,1}) + \varepsilon_{\iota,1}(\aleph_{\iota,1}) \tag{29}$$

where $\mathcal{Q}^*_{\iota,1} \in \mathbb{R}^{q_{\iota 1}}$ is the ideal neural network (NN) weight vector, $\mathfrak{W}_{\iota,1} \in \mathbb{R}^{q_{\iota 1}}$ is the basis function vector, and $\varepsilon_{\iota,1} \in \mathbb{R}$ is the approximation error.

Using (29) in (27) and (28), one has

$$\frac{d\mathcal{D}^*_{\iota,1}(\aleph_{\iota,1})}{d\aleph_{\iota,1}} = \frac{2\lambda_{\iota,1}\aleph_{\iota,1}(t)}{\Psi^{-1}_{\iota,1}(b_\iota + p_\iota)}\frac{1}{1-\aleph^2_{\iota,1}(t)} + \frac{2\wp_{\iota,1}(t)}{\Psi^{-1}_{\iota,1}(b_\iota + p_\iota)} + \frac{-2}{\Psi^{-1}_{\iota,1}(b_\iota + p_\iota)^2}\left(\sum_{l\in N_\iota} b_{\iota l}(\mathfrak{X}_{l,2}(t) + \wp_{l,1}(t)) + p_\iota \dot{y}_0(t)\right) + \frac{1-\aleph^2_{\iota,1}(t)}{\Psi^{-2}_{\iota,1}(b_\iota + p_\iota)^2}(\mathcal{Q}^{*T}_{\iota,1}\mathfrak{W}_{\iota,1}(\aleph_{\iota,1}) + \varepsilon_{\iota,1}(\aleph_{\iota,1})) \tag{30}$$

$$\alpha^*_{\iota,1}(t) = -\lambda_{\iota,1}\frac{\aleph_{\iota,1}(t)}{1-\aleph^2_{\iota,1}(t)} - \wp_{\iota,1}(t) + \frac{1}{b_\iota + p_\iota} \times \left(\sum_{l\in N_\iota} b_{\iota l}(\mathfrak{X}_{l,2}(t) + \wp_{l,1}(t)) + p_\iota \dot{y}_0(t)\right) - \frac{1-\aleph^2_{\iota,1}(t)}{2\Psi^{-1}_{\iota,1}(b_\iota + p_\iota)}(\mathcal{Q}^{*T}_{\iota,1}\mathfrak{W}_{\iota,1}(\aleph_{\iota,1}) + \varepsilon_{\iota,1}(\aleph_{\iota,1})) \tag{31}$$

Since $\mathcal{Q}^*_{\iota,1}$ is unknown, the optimal virtual control (31) is not available. Therefore, RL is adopted to design the critic and actor.

Based on (30) and (31), the critic and actor are designed to assess the control performance and generate the control action, respectively, as follows:

$$\frac{d\hat{\mathcal{D}}^*_{\iota,1}(\aleph_{\iota,1})}{d\aleph_{\iota,1}} = \frac{2\lambda_{\iota,1}\aleph_{\iota,1}(t)}{\Psi^{-1}_{\iota,1}(b_\iota + p_\iota)}\frac{1}{1-\aleph^2_{\iota,1}(t)} + \frac{2\wp_{\iota,1}(t)}{\Psi^{-1}_{\iota,1}(b_\iota + p_\iota)} + \frac{-2}{\Psi^{-1}_{\iota,1}(b_\iota + p_\iota)^2}\left(\sum_{l\in N_\iota} b_{\iota l}(\mathfrak{X}_{l,2}(t) + \wp_{l,1}(t)) + p_\iota \dot{y}_0(t)\right) + \frac{1-\aleph^2_{\iota,1}(t)}{\Psi^{-2}_{\iota,1}(b_\iota + p_\iota)^2}\hat{\mathcal{Q}}^T_{c,\iota 1}\mathfrak{W}_{\iota,1}(\aleph_{\iota,1})$$

and

$$\hat{\alpha}^*_{\iota,1}(t) = -\lambda_{\iota,1}\frac{\aleph_{\iota,1}(t)}{1-\aleph^2_{\iota,1}(t)} - \wp_{\iota,1}(t) + \frac{1}{b_\iota + p_\iota} \times \left(\sum_{l\in N_\iota} b_{\iota l}(\mathfrak{X}_{l,2}(t) + \wp_{l,1}(t)) + p_\iota \dot{y}_0(t)\right) - \frac{1-\aleph^2_{\iota,1}(t)}{2\Psi^{-1}_{\iota,1}(b_\iota + p_\iota)}\hat{\mathcal{Q}}^T_{a,\iota 1}\mathfrak{W}_{\iota,1}(\aleph_{\iota,1}) \tag{32}$$

where $(d\hat{\mathcal{D}}^*_{\iota,1}(\aleph_{\iota,1})/d\aleph_{\iota,1})$ and $\hat{\alpha}^*_{\iota,1}(t)$ is the estimation of $(d\mathcal{D}^*_{\iota,1}(\aleph_{\iota,1})/d\aleph_{\iota,1})$ and $\alpha^*_{\iota,1}(t)$, respectively.

The weights $\hat{\mathcal{Q}}_{c,\iota 1}$ and $\hat{\mathcal{Q}}_{a,\iota 1}$ are updated by

$$\dot{\hat{\mathcal{Q}}}_{c,\iota 1}(t) = -\beta_{c,\iota 1}\mathfrak{W}_{\iota,1}(\aleph_{\iota,1})\mathfrak{W}^T_{\iota,1}(\aleph_{\iota,1})\hat{\mathcal{Q}}_{c,\iota 1}(t) - \frac{1}{2}\mathfrak{W}_{\iota,1}(\aleph_{\iota,1})\aleph_{\iota,1}(t) \tag{33}$$

$$\dot{\hat{\mathcal{Q}}}_{a,\iota 1}(t) = -\beta_{a,\iota 1}\mathfrak{W}_{\iota,1}(\aleph_{\iota,1})\mathfrak{W}^T_{\iota,1}(\aleph_{\iota,1})(\hat{\mathcal{Q}}_{a,\iota 1}(t) - \hat{\mathcal{Q}}_{c,\iota 1}(t)) \tag{34}$$

where $\beta_{c,\iota 1} > 0$ and $\beta_{a,\iota 1} > 0$ are the designed parameters.

Consider the Lyapunov function candidate

$$V_{\iota,1}(t) = \frac{1}{2}\mathcal{K}_{\iota,1}(t) + \frac{1}{2}\tilde{\mathcal{Q}}^T_{c,\iota 1}(t)\tilde{\mathcal{Q}}_{c,\iota 1}(t) + \frac{1}{2}\tilde{\mathcal{Q}}^T_{a,\iota 1}(t)\tilde{\mathcal{Q}}_{a,\iota 1}(t)$$

where $\mathcal{K}_{\iota,1}(t) = \log\left(1/(1-\aleph^2_{\iota,1}(t))\right)$, $\tilde{\mathcal{Q}}_{c,\iota 1}(t) = \hat{\mathcal{Q}}_{c,\iota 1}(t) - \mathcal{Q}^*_{\iota,1}(t)$, and $\tilde{\mathcal{Q}}_{a,\iota 1}(t) = \hat{\mathcal{Q}}_{a,\iota 1}(t) - \mathcal{Q}^*_{\iota,1}(t)$; moreover, $\mathcal{K}_{\iota,1}(t) \geq 0$ holds for all $t \in [0, t_\tau)$ since $\aleph_{\iota,1}(t) \in (-1, 1)$.

Taking the time derivative of $V_{\iota,1}(t)$, along with (25), yields

$$\dot{V}_{\iota,1}(t) = \frac{\aleph_{\iota,1}(t)}{1-\aleph^2_{\iota,1}(t)}\Big[\Psi^{-1}_{\iota,1}(b_\iota + p_\iota)(\zeta_{\iota,2}(t) + \hat{\alpha}^*_{\iota,1}(t) + \wp_{\iota,1}(t) + \varrho_{\iota,1}(t)) - \sum_{l\in N_\iota} b_{\iota l}(\mathfrak{X}_{l,2}(t) + \wp_{l,1}(t) + \varrho_{l,1}(t)) - p_\iota \dot{y}_0(t) - \dot{\Psi}_{\iota,1}\aleph_{\iota,1}(t)\Big] + \tilde{\mathcal{Q}}^T_{c,\iota 1}(t)\dot{\hat{\mathcal{Q}}}_{c,\iota 1}(t) + \tilde{\mathcal{Q}}^T_{a,\iota 1}(t)\dot{\hat{\mathcal{Q}}}_{a,\iota 1}(t). \tag{35}$$

With $\tilde{\mathcal{Q}}_{c,\iota 1}(t) = \hat{\mathcal{Q}}_{c,\iota 1}(t) - \mathcal{Q}^*_{\iota 1}$ and $\tilde{\mathcal{Q}}_{a,\iota 1}(t) = \hat{\mathcal{Q}}_{a,\iota 1}(t) - \mathcal{Q}^*_{\iota 1}$, it follows that

$$-\frac{1}{2}\aleph_{\iota,1}(t)\hat{\mathcal{Q}}^T_{a,\iota 1}(t)\mathfrak{M}_{\iota,1}(\aleph_{\iota,1}) - \frac{1}{2}\tilde{\mathcal{Q}}^T_{c,\iota 1}(t)\mathfrak{M}_{\iota,1}(\aleph_{\iota,1})\aleph_{\iota,1}(t)$$

$$= -\frac{1}{2}\aleph_{\iota,1}(t)\tilde{\mathcal{Q}}_{a,\iota 1}^T(t)\mathfrak{M}_{\iota,1}(\aleph_{\iota,1}) - \frac{1}{2}\aleph_{\iota,1}(t)\hat{\mathcal{Q}}_{c,\iota 1}^T(t)\mathfrak{M}_{\iota,1}(\aleph_{\iota,1}) \tag{36}$$

$$\hat{\mathcal{Q}}_{a,\iota 1}(t) - \hat{\mathcal{Q}}_{c,\iota 1}(t) = \tilde{\mathcal{Q}}_{a,\iota 1}(t) - \tilde{\mathcal{Q}}_{c,\iota 1}(t). \tag{37}$$

By applying the optimized control (32)–(35) together with (33)–(34) and (36)–(37), one has

$$\begin{aligned}\dot{V}_{\iota,1}(t) = {}& \frac{\aleph_{\iota,1}(t)}{1-\aleph_{\iota,1}^2(t)}\Big[-\lambda_{\iota,1}\frac{\aleph_{\iota,1}(t)}{1-\aleph_{\iota,1}^2(t)}\Psi_{\iota,1}^{-1}(b_\iota+p_\iota)\\ &+\mathcal{F}_{\iota,1}(t)\Big] - \frac{1}{2}\aleph_{\iota,1}(t)\tilde{\mathcal{Q}}_{a,\iota 1}^T(t)\mathfrak{W}_{\iota,1}(\aleph_{\iota,1})\\ &- \frac{1}{2}\aleph_{\iota,1}(t)\hat{\mathcal{Q}}_{c,\iota 1}^T(t)\mathfrak{W}_{\iota,1}(\aleph_{\iota,1})\\ &- \beta_{c,\iota 1}\tilde{\mathcal{Q}}_{c,\iota 1}^T(t)\mathfrak{W}_{\iota,1}(\aleph_{\iota,1})\mathfrak{W}_{\iota,1}^T(\aleph_{\iota,1})\hat{\mathcal{Q}}_{c,\iota 1}(t)\\ &- \beta_{a,\iota 1}\tilde{\mathcal{Q}}_{a,\iota 1}^T(t)\mathfrak{W}_{\iota,1}(\aleph_{\iota,1})\mathfrak{W}_{\iota,1}^T(\aleph_{\iota,1})\tilde{\mathcal{Q}}_{a,\iota 1}(t)\\ &+ \beta_{a,\iota 1}\tilde{\mathcal{Q}}_{a,\iota 1}^T(t)\mathfrak{W}_{\iota,1}(\aleph_{\iota,1})\mathfrak{W}_{\iota,1}^T(\aleph_{\iota,1})\tilde{\mathcal{Q}}_{c,\iota 1}(t)\end{aligned}$$

where $\mathcal{F}_{\iota,1}(t) = \Psi_{\iota,1}^{-1}[(b_\iota+p_\iota)(\zeta_{\iota,2}+\varrho_{\iota,1}) - \sum_{l\in N_\iota} b_{\iota l}\varrho_{l,1} - \dot{\Psi}_{\iota,1}\aleph_{\iota,1}(t)]$.

According to the Young's inequality, the following result is obtained:

$$\begin{aligned}\dot{V}_{\iota,1}(t) \le {}& \frac{\aleph_{\iota,1}(t)}{1-\aleph_{\iota,1}^2(t)}\left[-\lambda_{\iota,1}\frac{\aleph_{\iota,1}(t)}{1-\aleph_{\iota,1}^2(t)}\Psi_{\iota,1}^{-1}(b_\iota+p_\iota)+\mathcal{F}_{\iota,1}(t)\right]\\ &-\left(\frac{\beta_{c,\iota 1}}{2}-\frac{\beta_{a,\iota 1}}{2}\right)\tilde{\mathcal{Q}}_{c,\iota 1}^T(t)\mathfrak{W}_{\iota,1}(\aleph_{\iota,1})\mathfrak{W}_{\iota,1}^T(\aleph_{\iota,1})\tilde{\mathcal{Q}}_{c,\iota 1}(t)\\ &-\left(\frac{\beta_{a,\iota 1}}{2}-\frac{1}{4}\right)\tilde{\mathcal{Q}}_{a,\iota 1}^T(t)\mathfrak{W}_{\iota,1}(\aleph_{\iota,1})\mathfrak{W}_{\iota,1}^T(\aleph_{\iota,1})\tilde{\mathcal{Q}}_{a,\iota 1}(t)\\ &-\left(\frac{\beta_{c,\iota 1}}{2}-\frac{1}{4}\right)\hat{\mathcal{Q}}_{c,\iota 1}^T(t)\mathfrak{W}_{\iota,1}(\aleph_{\iota,1})\mathfrak{W}_{\iota,1}^T(\aleph_{\iota,1})\hat{\mathcal{Q}}_{c,\iota 1}(t)\\ &+\frac{1}{2}\beta_{c,\iota 1}\|\mathcal{Q}_{\iota,1}^{*T}\mathfrak{W}_{\iota,1}(\aleph_{\iota,1})\|^2+\frac{1}{2}\aleph_{\iota,1}^2(t).\end{aligned} \tag{38}$$

Since $\aleph_{\iota,1}$ is bounded for all $t\in[0,t_\tau)$. Consequently, the boundedness of $\Psi_{\iota,2}$ implies that $\zeta_{\iota,2}$ are also bounded. Moreover, since $\varrho_{\iota,1}$ and $\varrho_{l,1}$ are bounded, there exists a positive constant $\bar{\mathcal{F}}_{\iota,1}$ such that $|\mathcal{F}_{\iota,1}|\le\bar{\mathcal{F}}_{\iota,1}$. Then (38) can be reformulated as follows:

$$\begin{aligned}\dot{V}_{\iota,1}(t) \le {}& \left(\frac{\epsilon_{\iota,1}}{2}-\lambda_{\iota,1}\underline{\Psi}_{\iota,1}^{-1}(b_\iota+p_\iota)\right)\frac{\aleph_{\iota,1}^2(t)}{(1-\aleph_{\iota,1}^2(t))^2}+\frac{1}{2\epsilon_{\iota,1}}\bar{\mathcal{F}}_{\iota,1}^2\\ &-\left(\frac{\beta_{c,\iota 1}}{2}-\frac{\beta_{a,\iota 1}}{2}\right)\lambda_{\mathfrak{W}_{\iota,1}}^{\min}\tilde{\mathcal{Q}}_{c,\iota 1}^T(t)\tilde{\mathcal{Q}}_{c,\iota 1}(t)\\ &-\left(\frac{\beta_{a,\iota 1}}{2}-\frac{1}{4}\right)\lambda_{\mathfrak{W}_{\iota,1}}^{\min}\tilde{\mathcal{Q}}_{a,\iota 1}^T(t)\tilde{\mathcal{Q}}_{a,\iota 1}(t)\\ &+\frac{1}{2}\beta_{c,\iota 1}\|\mathcal{Q}_{\iota,1}^{*T}\mathfrak{W}_{\iota,1}(\aleph_{\iota,1})\|^2+\frac{1}{2}\aleph_{\iota,1}^2(t)\end{aligned}$$

where $\underline{\Psi}_{\iota,1}^{-1}$ is a positive constant satisfying $\underline{\Psi}_{\iota,1}^{-1} < \left|\Psi_{\iota,1}^{-1}\right|$.

According to the fact that $-\log(1/1-\aleph_{\iota,1}^2(t)) \ge -\frac{\aleph_{\iota,1}^2(t)}{1-\aleph_{\iota,1}^2(t)}$, one has

$$\begin{aligned}\dot{V}_{\iota,1}(t) \le {}& \left(\frac{\epsilon_{\iota,1}}{2}-\lambda_{\iota,1}\underline{\Psi}_{\iota,1}^{-1}(b_\iota+p_\iota)\right)\frac{\aleph_{\iota,1}^2(t)}{(1-\aleph_{\iota,1}^2(t))^2}+\eta_{\iota,1}\\ &-\left(\frac{\beta_{c,\iota 1}}{2}-\frac{\beta_{a,\iota 1}}{2}\right)\lambda_{\mathfrak{W}_{\iota,1}}^{\min}\tilde{\mathcal{Q}}_{c,\iota 1}^T(t)\tilde{\mathcal{Q}}_{c,\iota 1}(t)\\ &-\left(\frac{\beta_{a,\iota 1}}{2}-\frac{1}{4}\right)\lambda_{\mathfrak{W}_{\iota,1}}^{\min}\tilde{\mathcal{Q}}_{a,\iota 1}^T(t)\tilde{\mathcal{Q}}_{a,\iota 1}(t)\\ \le {}& -a_{\iota,1}V_{\iota,1}(t)+\eta_{\iota,1}\end{aligned} \tag{39}$$

where $a_{\iota,1} = min\{(-\epsilon_{\iota,1} + 2\lambda_{\iota,1}\underline{\Psi}_{\iota,1}^{-1}(b_\iota+p_\iota)), (\beta_{c,\iota 1}-\beta_{a,\iota 1})\lambda_{\mathfrak{W}_{\iota,1}}^{\min}, (\beta_{a,\iota 1} - (1/2))\lambda_{\mathfrak{W}_{\iota,1}}^{\min}\}$, $\eta_{\iota,1} = (1/2)\beta_{c,\iota 1}\|\mathcal{Q}_{\iota,1}^{*T}\mathfrak{W}_{\iota,1}(\aleph_{\iota,1})\|^2 + (1/2)\bar{\aleph}_{\iota,1}^2$, $\beta_{a,\iota 1} > (1/2)$, and $\beta_{c,\iota 1} > \beta_{a,\iota 1}$.

Step $\ell(\ell = 2\ldots n-1)$: The optimal performance index can be written as follows:

$$\mathcal{D}_{\iota,\ell}^*(\aleph_{\iota,\ell}) = \int_t^\infty c_{\iota,\ell}(\aleph_{\iota,\ell}(s),\alpha_{\iota,\ell}^*(s))ds. \tag{40}$$

By solving $\partial H_{\iota,\ell}/\partial\alpha_{\iota,\ell}^* = 0$, one has

$$\alpha_{\iota,\ell}^*(t) = -\frac{1}{2}\Psi_{\iota,\ell}^{-1}\frac{d\mathcal{D}_{\iota,\ell}^*(\aleph_{\iota,\ell})}{d\aleph_{\iota,\ell}}.$$

Then, reconstruct $(d\mathcal{D}_{\iota,\ell}^*(\aleph_{\iota,\ell})/d\aleph_{\iota,\ell})$ as

$$\frac{d\mathcal{D}_{\iota,\ell}^*(\aleph_{\iota,\ell})}{d\aleph_{\iota,\ell}} = \frac{2\lambda_{\iota,\ell}\aleph_{\iota,\ell}(t)}{\Psi_{\iota,\ell}^{-1}(1-\aleph_{\iota,\ell}^2(t))} - \frac{2}{\Psi_{\iota,\ell}^{-1}}(-\wp_{\iota,\ell}(\bar{\mathfrak{X}}_{\iota,\ell}) + \dot{\hat{\alpha}}_{\iota,\ell-1}^*(t)) + \frac{1-\aleph_{\iota,\ell}^2(t)}{\Psi_{\iota,\ell}^{-2}}\mathcal{D}_{\iota,\ell}^o$$

where $\lambda_{\iota,\ell} > 0$ is a constant, and $\mathcal{D}_{\iota,\ell}^o = (2\Psi_{\iota,\ell}^{-1}/1-\aleph_{\iota,\ell}^2(t))(-\lambda_{\iota,\ell}(\aleph_{\iota,\ell}(t)/1-\aleph_{\iota,\ell}^2(t)) - \wp_{\iota,\ell}(\bar{\mathfrak{X}}_{\iota,\ell}) + \dot{\hat{\alpha}}_{\iota,\ell-1}^*(t)) + (\Psi_{\iota,\ell}^{-2}/1-\aleph_{\iota,\ell}^2)(d\mathcal{D}_{\iota,\ell}^*(\aleph_{\iota,\ell})/d\aleph_{\iota,\ell})$.

Then the optimal virtual controller $\alpha_{\iota,\ell}^*(t)$ can be re-expressed as follows:

$$\alpha_{\iota,\ell}^*(t) = -\lambda_{\iota,\ell}\frac{\aleph_{\iota,\ell}(t)}{1-\aleph_{\iota,\ell}^2(t)} - \wp_{\iota,\ell}(t) + \dot{\hat{\alpha}}_{\iota,\ell-1}^*(t) - \frac{1-\aleph_{\iota,\ell}^2(t)}{2\Psi_{\iota,\ell}^{-1}(t)}\mathcal{D}_{\iota,\ell}^o.$$

Then the unknown continuous term $\mathcal{D}_{\iota,\ell}^o$ can be approximated by a neural network over a compact set as follows:

$$\mathcal{D}_{\iota,\ell}^o = \mathcal{Q}_{\iota,\ell}^{*T}\mathfrak{W}_{\iota,\ell}(\aleph_{\iota,\ell}) + \varepsilon_{\iota,\ell}(\aleph_{\iota,\ell})$$

where $\mathcal{Q}_{\iota,\ell}^* \in \mathbb{R}^{q_{\iota\ell}}$ is the ideal NN weight vector, $\mathfrak{W}_{\iota,\ell}\in\mathbb{R}^{q_{\iota\ell}}$ is the basis function vector, and $\varepsilon_{\iota,\ell}\in\mathbb{R}$ is the approximation error.

In this framework, the critic and actor are constructed to evaluate the control performance and generate the control action, respectively, as follows:

$$\frac{d\hat{\mathcal{D}}_{\iota,\ell}^*(\aleph_{\iota,\ell})}{d\aleph_{\iota,\ell}} = \frac{2\lambda_{\iota,\ell}\aleph_{\iota,\ell}(t)}{\Psi_{\iota,\ell}^{-1}(1-\aleph_{\iota,\ell}^2(t))} - \frac{2}{\Psi_{\iota,\ell}^{-1}}(-\wp_{\iota,\ell}(t) + \dot{\hat{\alpha}}_{\iota,\ell-1}^*(t)) + \frac{1-\aleph_{\iota,\ell}^2(t)}{2\Psi_{\iota,\ell}^{-1}}\hat{\mathcal{Q}}_{c,\iota\ell}^T\mathfrak{W}_{\iota,\ell}(\aleph_{\iota,\ell})$$

and

$$\hat{\alpha}_{\iota,\ell}^*(t) = -\lambda_{\iota,\ell}\frac{\aleph_{\iota,\ell}(t)}{1-\aleph_{\iota,\ell}^2(t)} - \wp_{\iota,\ell}(t) + \dot{\hat{\alpha}}_{\iota,\ell-1}^*(t)$$

$$-\frac{1-\aleph_{\iota,\ell}^2(t)}{2\Psi_{\iota,\ell}^{-1}(t)}\hat{\mathcal{Q}}_{a,\iota\ell}^T\mathfrak{W}_{\iota,\ell}(\aleph_{\iota,\ell}) \tag{41}$$

where $(d\hat{\mathcal{D}}_{\iota,\ell}^*(\aleph_{\iota,\ell})/d\aleph_{\iota,\ell})$ and $\hat{\alpha}_{\iota,\ell}^*(t)$ is the estimation of $(d\mathcal{D}_{\iota,\ell}^*(\aleph_{\iota,\ell})/d\aleph_{\iota,\ell})$ and $\alpha_{\iota,\ell}^*(t)$, respectively.

The update laws for the critic weight $\hat{\mathcal{Q}}_{c,\iota\ell}$ and the actor weight $\hat{\mathcal{Q}}_{a,\iota\ell}$ are given by

$$\dot{\hat{\mathcal{Q}}}_{c,\iota\ell}(t) = -\beta_{c,\iota\ell}\mathfrak{W}_{\iota,\ell}(\aleph_{\iota,\ell})\mathfrak{W}_{\iota,\ell}^T(\aleph_{\iota,\ell})\hat{\mathcal{Q}}_{c,\iota\ell}(t) - \frac{1}{2}\mathfrak{W}_{\iota,\ell}(\aleph_{\iota,\ell})\aleph_{\iota,\ell}(t) \tag{42}$$

$$\dot{\hat{\mathcal{Q}}}_{a,\iota\ell}(t) = -\beta_{a,\iota\ell}\mathfrak{W}_{\iota,\ell}(\aleph_{\iota,\ell})\mathfrak{W}_{\iota,\ell}^T(\aleph_{\iota,\ell}) \times (\hat{\mathcal{Q}}_{a,\iota\ell}(t) - \hat{\mathcal{Q}}_{c,\iota\ell}(t)) \tag{43}$$

where $\beta_{c,\iota\ell} > 0$ and $\beta_{a,\iota\ell} > 0$ are the designed parameter.

The Lyapunov function candidate is chosen as follows:

$$V_{\iota,\ell}(t) = \frac{1}{2}\mathcal{K}_{\iota,\ell}(t) + \frac{1}{2}\tilde{\mathcal{Q}}_{c,\iota\ell}^T(t)\tilde{\mathcal{Q}}_{c,\iota\ell}(t) + \frac{1}{2}\tilde{\mathcal{Q}}_{a,\iota\ell}^T(t)\tilde{\mathcal{Q}}_{a,\iota\ell}(t) \tag{44}$$

where $\mathcal{K}_{\iota,\ell}(t) = \log\left(1/(1-\aleph_{\iota,\ell}^2(t))\right)$, $\tilde{\mathcal{Q}}_{c,\iota\ell}(t) = \hat{\mathcal{Q}}_{c,\iota\ell}(t) - \mathcal{Q}_{\iota,\ell}^*(t)$ and $\tilde{\mathcal{Q}}_{a,\iota\ell}(t) = \hat{\mathcal{Q}}_{a,\iota\ell}(t) - \mathcal{Q}_{\iota,\ell}^*(t)$.

Similar to (36) and (37), one can get

$$-\frac{1}{2}\aleph_{\iota,\ell}(t)\hat{\mathcal{Q}}_{a,\iota\ell}^T(t)\mathfrak{M}_{\iota,\ell}(\aleph_{\iota,\ell}) - \frac{1}{2}\tilde{\mathcal{Q}}_{c,\iota\ell}^T(t)\mathfrak{M}_{\iota,\ell}(\aleph_{\iota,\ell})\aleph_{\iota,\ell}(t) = -\frac{1}{2}\aleph_{\iota,\ell}(t)\tilde{\mathcal{Q}}_{a,\iota\ell}^T(t)\mathfrak{M}_{\iota,\ell}(\aleph_{\iota,\ell}) - \frac{1}{2}\aleph_{\iota,\ell}(t)\hat{\mathcal{Q}}_{c,\iota\ell}^T(t)\mathfrak{M}_{\iota,\ell}(\aleph_{\iota,\ell}) \tag{45}$$

$$\hat{\mathcal{Q}}_{a,\iota\ell}(t) - \hat{\mathcal{Q}}_{c,\iota\ell}(t) = \tilde{\mathcal{Q}}_{a,\iota\ell}(t) - \tilde{\mathcal{Q}}_{c,\iota\ell}(t). \tag{46}$$

By applying the optimized control (41)–(44) together with (42)–(43) and (45)–(46), one has

$$\begin{aligned}\dot{V}_{\iota,\ell}(t) = &\frac{\aleph_{\iota,\ell}(t)}{1-\aleph_{\iota,\ell}^2(t)}\left[-\lambda_{\iota,\ell}\frac{\aleph_{\iota,\ell}(t)}{1-\aleph_{\iota,\ell}^2(t)}\Psi_{\iota,\ell}^{-1}(t) + \mathcal{F}_{\iota,\ell}(t)\right] \\ &- \frac{1}{2}\aleph_{\iota,\ell}(t)\tilde{\mathcal{Q}}_{a,\iota\ell}^T(t)\mathfrak{W}_{\iota,\ell}(\aleph_{\iota,\ell}) \\ &- \frac{1}{2}\aleph_{\iota,\ell}(t)\hat{\mathcal{Q}}_{c,\iota\ell}^T(t)\mathfrak{W}_{\iota,\ell}(\aleph_{\iota,\ell}) \\ &- \beta_{c,\iota\ell}\tilde{\mathcal{Q}}_{c,\iota\ell}^T(t)\mathfrak{W}_{\iota,\ell}(\aleph_{\iota,\ell})\mathfrak{W}_{\iota,\ell}^T(\aleph_{\iota,\ell})\hat{\mathcal{Q}}_{c,\iota\ell}(t) \\ &- \beta_{a,\iota\ell}\tilde{\mathcal{Q}}_{a,\iota\ell}^T(t)\mathfrak{W}_{\iota,\ell}(\aleph_{\iota,\ell})\mathfrak{W}_{\iota,\ell}^T(\aleph_{\iota,\ell})\tilde{\mathcal{Q}}_{a,\iota\ell}(t) \\ &+ \beta_{a,\iota\ell}\tilde{\mathcal{Q}}_{a,\iota\ell}^T(t)\mathfrak{W}_{\iota,\ell}(\aleph_{\iota,\ell})\mathfrak{W}_{\iota,\ell}^T(\aleph_{\iota,\ell})\tilde{\mathcal{Q}}_{c,\iota\ell}(t)\end{aligned}$$

where $\mathcal{F}_{\iota,\ell}(t) = \Psi_{\iota,\ell}^{-1}(t)(\zeta_{\iota,\ell+1} + \varrho_{\iota,\ell} - \dot{\Psi}_{\iota,\ell}(t)\aleph_{\iota,\ell}(t))$, and there exists a positive constant $\bar{\mathcal{F}}_{\iota,\ell}$ such that $|\mathcal{F}_{\iota,\ell}| \le \bar{\mathcal{F}}_{\iota,\ell}$.

Similar to (38) and (39), the following result is obtained:

$$\begin{aligned}\dot{V}_{\iota,\ell}(t) \le &\frac{\aleph_{\iota,\ell}(t)}{1-\aleph_{\iota,\ell}^2(t)}\left[-\lambda_{\iota,\ell}\frac{\aleph_{\iota,\ell}(t)}{1-\aleph_{\iota,\ell}^2(t)}\Psi_{\iota,\ell}^{-1}(t) + \mathcal{F}_{\iota,\ell}(t)\right] \\ &- \left(\frac{\beta_{c,\iota\ell}}{2} - \frac{\beta_{a,\iota\ell}}{2}\right)\tilde{\mathcal{Q}}_{c,\iota\ell}^T(t)\mathfrak{W}_{\iota,\ell}(\aleph_{\iota,\ell})\mathfrak{W}_{\iota,\ell}^T(\aleph_{\iota,\ell})\tilde{\mathcal{Q}}_{c,\iota\ell}(t) \\ &- \left(\frac{\beta_{a,\iota\ell}}{2} - \frac{1}{4}\right)\tilde{\mathcal{Q}}_{a,\iota\ell}^T(t)\mathfrak{W}_{\iota,1}(\aleph_{\iota,\ell})\mathfrak{W}_{\iota,\ell}^T(\aleph_{\iota,\ell})\tilde{\mathcal{Q}}_{a,\iota\ell}(t) \\ &- \left(\frac{\beta_{c,\iota\ell}}{2} - \frac{1}{4}\right)\hat{\mathcal{Q}}_{c,\iota\ell}^T(t)\mathfrak{W}_{\iota,\ell}(\aleph_{\iota,\ell})\mathfrak{W}_{\iota,\ell}^T(\aleph_{\iota,\ell})\hat{\mathcal{Q}}_{c,\iota\ell}(t) \\ &+ \frac{1}{2}\beta_{c,\iota\ell}\|\mathcal{Q}_{\iota,\ell}^{*T}\mathfrak{W}_{\iota,\ell}(\aleph_{\iota,\ell})\|^2 + \frac{1}{2}\aleph_{\iota,\ell}^2(t) \end{aligned} \tag{47}$$

$$\begin{aligned} \le &\left(\frac{\epsilon_{\iota,\ell}}{2} - \lambda_{\iota,\ell}\underline{\Psi}_{\iota,\ell}^{-1}\right)\frac{\aleph_{\iota,\ell}^2(t)}{(1-\aleph_{\iota,\ell}^2(t))^2} + \frac{1}{2\epsilon_{\iota,\ell}}\bar{\mathcal{F}}_{\iota,\ell}^2 \\ &- \left(\frac{\beta_{c,\iota\ell}}{2} - \frac{\beta_{a,\iota\ell}}{2}\right)\lambda_{\mathfrak{W}_{\iota,\ell}}^{\min}\tilde{\mathcal{Q}}_{c,\iota\ell}^T(t)\tilde{\mathcal{Q}}_{c,\iota\ell}(t) \\ &- \left(\frac{\beta_{a,\iota\ell}}{2} - \frac{1}{4}\right)\lambda_{\mathfrak{W}_{\iota,\ell}}^{\min}\tilde{\mathcal{Q}}_{a,\iota\ell}^T(t)\tilde{\mathcal{Q}}_{a,\iota\ell}(t) \\ &+ \frac{1}{2}\beta_{c,\iota\ell}\|\mathcal{Q}_{\iota,\ell}^{*T}\mathfrak{W}_{\iota,\ell}(\aleph_{\iota,\ell})\|^2 + \frac{1}{2}\aleph_{\iota,\ell}^2(t) \\ \le &-a_{\iota,\ell}V_{\iota,\ell}(t) + \eta_{\iota,\ell}\end{aligned} \tag{48}$$

where $a_{\iota,\ell} = min\{(-\epsilon_{\iota,\ell} + 2\lambda_{\iota,\ell}\underline{\Psi}_{\iota,\ell}^{-1}), (\beta_{c,\iota\ell} - \beta_{a,\iota\ell})\lambda_{\mathfrak{W}_{\iota,\ell}}^{\min}, (\beta_{a,\iota\ell} - (1/2)\lambda_{\mathfrak{W}_{\iota,\ell}}^{\min}\}, \eta_{\iota,\ell} = (1/2)\beta_{c,\iota\ell}\|\mathcal{Q}_{\iota,\ell}^{*T}\mathfrak{W}_{\iota,\ell}(\aleph_{\iota,\ell})\|^2 + (1/2)\bar{\aleph}_{\iota,\ell}^2, \beta_{a,\iota\ell} > (1/2)$, and $\beta_{c,\iota\ell} > \beta_{a,\iota\ell}$.

Step n: Define the following indicator functions as:

$$\mathcal{I}_{\iota,1}(t) = \begin{cases} 1, & \text{if } t \ge t_{\iota,f}, \text{ i.e., fault occurs} \\ 0, & \text{otherwise}\end{cases}$$

$$\text{and}\quad \mathcal{I}_{\iota,2}(t) = \begin{cases} 1, & \text{if } t \ge \hat{t}_{\iota,f} \\ 0, & \text{otherwise}\end{cases}$$

where $t_{\iota,f}$ denotes the time at which the fault $\mathfrak{U}_{\iota,f}$ occurs, and $\hat{t}_{\iota,f}$ represents the time at which the fault is detected by (6). Define the FE error as $\tilde{\mathfrak{U}}_{\iota,f} = \mathcal{I}_{\iota,1}\mathfrak{U}_{\iota,f} - \mathcal{I}_{\iota,2}\hat{\mathfrak{U}}_{\iota,f}$.

Let $\mathfrak{U}_{\iota,0}^*(t)$ denote the optimal actual control, and the corresponding optimal performance index is given by

$$\mathcal{D}_{\iota,n}^*(\aleph_{\iota,n}) = \int_t^\infty c_{\iota,n}(\aleph_{\iota,n}(s), \mathfrak{U}_{\iota,0}^*(s))ds. \tag{49}$$

By solving $\partial H_{\iota,n}/\partial\mathfrak{U}_{\iota,0}^* = 0$, one has

$$\mathfrak{U}_{\iota,0}^*(t) = -\frac{b_\iota}{2}\Psi_{\iota,n}^{-1}\frac{d\mathcal{D}_{\iota,n}^*(\aleph_{\iota,n})}{d\aleph_{\iota,n}}. \tag{50}$$

Then, reconstruct $(d\mathcal{D}_{\iota,n}^*(\aleph_{\iota,n})/d\aleph_{\iota,n})$ as follows:

$$\begin{aligned}\frac{d\mathcal{D}_{\iota,n}^*(\aleph_{\iota,n})}{d\aleph_{\iota,n}} = &\frac{2\lambda_{\iota,n}\aleph_{\iota,n}(t)}{b_\iota^2\Psi_{\iota,n}^{-1}(1-\aleph_{\iota,n}^2(t))} - \frac{2}{b_\iota^2\Psi_{\iota,n}^{-1}} \\ &\times(-b_\iota\mathcal{I}_{\iota,2}\hat{\mathfrak{U}}_{\iota,f}(t) - \wp_{\iota,n}(\mathfrak{X}_{\iota,n}) + \dot{\hat{\alpha}}_{\iota,n-1}^*(t)) \\ &+ \frac{1-\aleph_{\iota,n}^2(t)}{b_\iota^2\Psi_{\iota,n}^{-2}}\mathcal{D}_{\iota,n}^o\end{aligned} \tag{51}$$

where $\lambda_{\iota,n} > 0$ is a constant, and $\mathcal{D}_{\iota,n}^o = (2\Psi_{\iota,n}^{-1}/1-\aleph_{\iota,n}^2(t))(-\lambda_{\iota,n}(\aleph_{\iota,n}(t)/1-\aleph_{\iota,n}^2(t)) - \wp_{\iota,n}(\mathfrak{X}_{\iota,n}) + \dot{\hat{\alpha}}_{\iota,n-1}^*(t)) + (b_\iota^2\Psi_{\iota,n}^{-2}/1-\aleph_{\iota,n}^2(t))(d\mathcal{D}_{\iota,n}^*(\aleph_{\iota,n})/d\aleph_{\iota,n})$.

Substituting (51) into (50) has

$$\mathfrak{U}_{\iota,0}^*(t) = \frac{1}{b_\iota}\left[-\lambda_{\iota,n}\frac{\aleph_{\iota,n}(t)}{1-\aleph_{\iota,n}^2(t)} - b_\iota\mathcal{I}_{\iota,2}\hat{\mathfrak{U}}_{\iota,f} - \wp_{\iota,n}(\mathfrak{X}_{\iota,n}) + \dot{\hat{\alpha}}_{\iota,n-1}^*(t) - \frac{1-\aleph_{\iota,n}^2(t)}{2\Psi_{\iota,n}^{-1}(t)}\mathcal{D}_{\iota,n}^o\right].$$

The unknown continuous term $\mathcal{D}^o_{\iota,n}$ is approximated by a neural network on a compact set as follows:

$$\mathcal{D}^o_{\iota,n} = \mathcal{Q}^{*T}_{\iota,n}\mathfrak{W}_{\iota,n}(\aleph_{\iota,n}) + \varepsilon_{\iota,n}(\aleph_{\iota,n})$$

where $\mathcal{Q}^*_{\iota,n}$ is the ideal weight vector, $\mathfrak{W}_{\iota,n}$ represents the basis function vector, and $\varepsilon_{\iota,n}$ is the approximation error.

Here, the critic evaluates performance and the actor generates the control action, as follows:

$$\begin{aligned}\frac{d\hat{\mathcal{D}}^*_{\iota,n}(\aleph_{\iota,n})}{d\aleph_{\iota,n}} &= \frac{2\lambda_{\iota,n}\aleph_{\iota,n}(t)}{b_\iota^2\Psi^{-1}_{\iota,n}(1-\aleph^2_{\iota,n}(t))} - \frac{2}{b_\iota^2\Psi^{-1}_{\iota,n}}\\ &\quad\times(-b_\iota\mathcal{I}_{\iota,2}\hat{\mathfrak{U}}_{\iota,f}(t) - \wp_{\iota,n}(\mathfrak{X}_{\iota,n}) + \dot{\alpha}^*_{\iota,n-1}(t))\\ &\quad+\frac{1-\aleph^2_{\iota,n}(t)}{b_\iota^2\Psi^{-2}_{\iota,n}}\hat{\mathcal{Q}}^T_{c,\iota n}\mathfrak{W}_{\iota,n}(\aleph_{\iota,n})\end{aligned}$$

and

$$\begin{aligned}\hat{\mathfrak{U}}^*_{\iota,0}(t) &= \frac{1}{b_\iota}\Big[-\lambda_{\iota,n}\frac{\aleph_{\iota,n}(t)}{1-\aleph^2_{\iota,n}(t)} - b_\iota\mathcal{I}_{\iota,2}\hat{\mathfrak{U}}_{\iota,f} - \wp_{\iota,n}(\mathfrak{X}_{\iota,n})\\ &\quad+\dot{\alpha}^*_{\iota,n-1}(t) - \frac{1-\aleph^2_{\iota,n}(t)}{2\Psi^{-1}_{\iota,n}(t)}\hat{\mathcal{Q}}^T_{a,\iota n}\mathfrak{W}_{\iota,n}(\aleph_{\iota,n})\Big]. \quad (52)\end{aligned}$$

The critic weight $\hat{\mathcal{Q}}_{c,\iota n}$ and the actor weight $\hat{\mathcal{Q}}_{a,\iota n}$ are updated according to

$$\begin{aligned}\dot{\hat{\mathcal{Q}}}_{c,\iota n}(t) &= -\beta_{c,\iota n}\mathfrak{W}_{\iota,n}(\aleph_{\iota,n})\mathfrak{W}^T_{\iota,n}(\aleph_{\iota,n})\hat{\mathcal{Q}}_{c,\iota n}(t)\\ &\quad-\frac{1}{2}\mathfrak{W}_{\iota,n}(\aleph_{\iota,n})\aleph_{\iota,n}(t) \quad (53)\\ \dot{\hat{\mathcal{Q}}}_{a,\iota n}(t) &= -\beta_{a,\iota n}\mathfrak{W}_{\iota,n}(\aleph_{\iota,n})\mathfrak{W}^T_{\iota,n}(\aleph_{\iota,n})\\ &\quad\times(\hat{\mathcal{Q}}_{a,\iota n}(t) - \hat{\mathcal{Q}}_{c,\iota n}(t)) \quad (54)\end{aligned}$$

where $\beta_{c,\iota n} > 0$ and $\beta_{a,\iota n} > 0$ are the designed parameter.

Select the following Lyapunov function candidate:

$$\begin{aligned}V_{\iota,n}(t) &= \frac{1}{2}\mathcal{K}_{\iota,n}(t) + \frac{1}{2}\tilde{\mathcal{Q}}^T_{c,\iota n}(t)\tilde{\mathcal{Q}}_{c,\iota n}(t)\\ &\quad+\frac{1}{2}\tilde{\mathcal{Q}}^T_{a,\iota n}(t)\tilde{\mathcal{Q}}_{a,\iota n}(t). \quad (55)\end{aligned}$$

Similar to the previous step, there are the following equations:

$$\begin{aligned}&-\frac{1}{2}\aleph_{\iota,n}(t)\hat{\mathcal{Q}}^T_{a,\iota n}(t)\mathfrak{M}_{\iota,n}(\aleph_{\iota,n}) - \frac{1}{2}\tilde{\mathcal{Q}}^T_{c,\iota n}(t)\mathfrak{M}_{\iota,n}(\aleph_{\iota,n})\aleph_{\iota,n}(t)\\ &\quad= -\frac{1}{2}\aleph_{\iota,n}(t)\tilde{\mathcal{Q}}^T_{a,\iota n}(t)\mathfrak{M}_{\iota,n}(\aleph_{\iota,n})\\ &\quad\quad-\frac{1}{2}\aleph_{\iota,n}(t)\hat{\mathcal{Q}}^T_{c,\iota n}(t)\mathfrak{M}_{\iota,n}(\aleph_{\iota,n}) \quad (56)\\ &\quad\hat{\mathcal{Q}}_{a,\iota n}(t) - \hat{\mathcal{Q}}_{c,\iota n}(t) = \tilde{\mathcal{Q}}_{a,\iota n}(t) - \tilde{\mathcal{Q}}_{c,\iota n}(t). \quad (57)\end{aligned}$$

Applying (52)–(55), using (53), (54), (56), and (57), and invoking Young's inequality yields

$$\begin{aligned}\dot{V}_{\iota,n}(t) &\le \left(\frac{\epsilon_{\iota,n}}{2} - \lambda_{\iota,n}\underline{\Psi}^{-1}_{\iota,n}\right)\frac{\aleph^2_{\iota,n}(t)}{(1-\aleph^2_{\iota,n}(t))^2} + \frac{1}{2\epsilon_{\iota,n}}\bar{\mathcal{F}}^2_{\iota,n}\\ &\quad-\left(\frac{\beta_{c,\iota n}}{2} - \frac{\beta_{a,\iota n}}{2}\right)\lambda^{\min}_{\mathfrak{W}_{\iota,n}}\tilde{\mathcal{Q}}^T_{c,\iota n}(t)\tilde{\mathcal{Q}}_{c,\iota n}(t)\\ &\quad-\left(\frac{\beta_{a,\iota n}}{2} - \frac{1}{4}\right)\lambda^{\min}_{\mathfrak{W}_{\iota,n}}\tilde{\mathcal{Q}}^T_{a,\iota n}(t)\tilde{\mathcal{Q}}_{a,\iota n}(t)\\ &\quad+\frac{1}{2}\beta_{c,\iota n}\|\mathcal{Q}^{*T}_{\iota,n}\mathfrak{W}_{\iota,n}(\aleph_{\iota,n})\|^2 + \frac{1}{2}\aleph^2_{\iota,n}(t),\\ &\le -a_{\iota,n}V_{\iota,n}(t) + \eta_{\iota,n} \quad (58)\end{aligned}$$

where $\mathcal{F}_{\iota,n}(t) = \Psi^{-1}_{\iota,n}(t)(b_\iota\tilde{\mathfrak{U}}_{\iota,f} + \varrho_{\iota,n} - \dot{\Psi}_{\iota,n}(t)\aleph_{\iota,n}(t))$, and there exists a positive constant $\bar{\mathcal{F}}_{\iota,n}$ such that $|\mathcal{F}_{\iota,n}| \le \bar{\mathcal{F}}_{\iota,n}$, $a_{\iota,n} = min\{(-\epsilon_{\iota,n} + 2\lambda_{\iota,n}\underline{\Psi}^{-1}_{\iota,n}), (\beta_{c,\iota n} - \beta_{a,\iota n})\lambda^{\min}_{\mathfrak{W}_{\iota,n}}, (\beta_{a,\iota n} - (1/2))\lambda^{\min}_{\mathfrak{W}_{\iota,n}}\}$, $\eta_{\iota,n} = (1/2)\beta_{c,\iota n}\|\mathcal{Q}^{*T}_{\iota,n}\mathfrak{W}_{\iota,n}(\aleph_{\iota,n})\|^2 + (1/2)\bar{\aleph}^2_{\iota,n}$, $\beta_{a,\iota n} > (1/2)$, $\beta_{c,\iota n} > \beta_{a,\iota n}$.

Finally, the Lyapunov function candidates are combined as

$$V(t) = \sum_{\iota=1}^{N}\sum_{\ell=1}^{n}V_{\iota,\ell}(t). \quad (59)$$

From (39), (48), and (58), the time derivative of (59) satisfies

$$\dot{V}(t) \le -aV(t) + \eta$$

where $a = \min\{a_{\iota,\ell}, \iota = 1,\ldots,N; \ell = 1,\ldots,n\}$, and $\eta = \sum_{\iota=1}^{N}\sum_{\ell=1}^{n}\eta_{\iota,\ell}$.

Hence, all closed-loop signals are bounded on $t \in [0, t_\tau]$. Then, there exists a compact set

$$\Lambda' = \prod_{\iota=1}^{N}\prod_{\ell=1}^{n}\big[-\bar{\aleph}_{\iota,\ell}, \bar{\aleph}_{\iota,\ell}\big] \subset \Lambda_\aleph$$

such that $\aleph(t) \in \Lambda'$ for all $t \in [0, t_\tau)$.

By applying a standard contradiction argument based on Lemma (3), the maximal solution is extendable to $t_\tau = +\infty$, which ensures $\aleph(t) \in \Lambda_\aleph$ and uniform boundedness of all closed-loop signals for all $t \ge 0$. Note that $|\aleph_{\iota,\ell}(t)| < 1$ for all $t \in [0, +\infty)$ directly yields $|\zeta_{\iota,\ell}(t)| < \Psi_{\iota,\ell}(t)$, which ensures that the funnel performance requirement is satisfied. ■

Remark 2: The proof of Theorem (3) is structured in three key stages. First, it establishes the existence and uniqueness of a maximal solution $\aleph(t)$ within the open set $\Lambda_\aleph$ for $t \in [0, t_\tau)$. Second, it demonstrates the boundedness of all closed-loop signals while $\aleph(t)$ constrained to a compact subset $\Lambda' \subset \Lambda_\aleph$. Finally, employing a contradiction argument via Lemma (3), the solution is proven to extend to $t \in [0, \infty)$, ensuring that the system stability and the fulfillment of the funnel performance condition for all time.

Remark 3: The actor-critic framework in RL is instrumental in deriving the optimal control policy. The critic network evaluates the control performance, while the actor network adjusts the control actions based on this evaluation. The term $\mathcal{D}^o$, representing the complex unknown nonlinear dynamics arising from the HJB equation, is approximated using a neural network whose weights $\mathcal{Q}^*$ learned through RL update laws.

B. Safety Analysis Based on CBF

To formalize the safety guarantee, we define a continuously differentiable function $\mathcal{E}$

$$\mathcal{E}(\mathfrak{X}_1,\ldots,\mathfrak{X}_n) = \sum_{\iota=1}^{N}\sum_{\ell=1}^{n}\frac{1}{2}\left(1-\aleph^2_{\iota,\ell}(t)\right) \quad (60)$$

where $\mathfrak{X}_\ell = [\mathfrak{X}_{1,\ell}, \ldots, \mathfrak{X}_{N,\ell}]^T$ for $\ell = 1, \ldots, n$. Safety is then characterized by the super-level set $\mathcal{G}$ of $\mathcal{E}$, defined as follows:

$$\mathcal{G} = \{(\mathfrak{X}_1, \ldots, \mathfrak{X}_n) \mid \mathcal{E}(\mathfrak{X}_1, \ldots, \mathfrak{X}_n) \geq 0\}.$$

Theorem 4: Under the Assumptions 1-4, the MASs (4) operating with the proposed funnel-based distributed controller preserves safety as specified in Definition (1). Specifically, the function $\mathcal{E}$ in (60) serves as a valid CBF on $\mathcal{G}$, thereby ensuring that system trajectories starting in the safe set $\mathcal{G}$ remain within it for all subsequent time.

Proof: According to Definition (1), the forward invariance of $\mathcal{G}$ is guaranteed if the following condition holds:

$$\sum_{k=1}^{n} \frac{\partial \mathcal{E}}{\partial \mathfrak{X}_k}(\mathfrak{X}_k)\dot{\mathfrak{X}}_k \geq -\varpi\left(\mathcal{E}(\mathfrak{X}_1, \ldots, \mathfrak{X}_n)\right) \tag{61}$$

where $\varpi(\cdot) \in \mathcal{K}_\infty^e$.

Step 1: Define a continuously differentiable function

$$\mathcal{E}_{\iota,1}(\mathfrak{X}_{\iota,1}) = \frac{1}{2}\left(1 - \aleph_{\iota,1}^2(t)\right).$$

From (25), one can derive the expression for $\dot{\mathcal{E}}_{\iota,1}(\mathfrak{X}_{\iota,1})$

$$\begin{aligned}\dot{\mathcal{E}}_{\iota,1}(\mathfrak{X}_{\iota,1}) &= -\aleph_{\iota,1}(t)\dot{\aleph}_{\iota,1}(t),\\ &= -\aleph_{\iota,1}(t)[-\lambda_{\iota,1}\frac{\aleph_{\iota,1}(t)}{1-\aleph_{\iota,1}^2(t)}\Psi_{\iota,1}^{-1}(b_\iota + p_\iota)\\ &\quad + \mathcal{H}_{\iota,1}(t)]\end{aligned}$$

where $\mathcal{H}_{\iota,1}(t) = \Psi_{\iota,1}^{-1}(t)[(b_\iota + p_\iota)(\zeta_{\iota,2}(t) - (1 - \aleph_{\iota,1}^2(t)/2\Psi_{\iota,1}^{-1}(t)(b_\iota + p_\iota))\hat{\mathcal{Q}}_{a,\iota 1}^T\mathfrak{W}_{\iota,1}(\aleph_{\iota,1}) + \varrho_{\iota,1}(t)) - \sum_{l\in N_\iota} b_{\iota l}\varrho_{l,1} - \dot{\Psi}_{\iota,1}\aleph_{\iota,1}(t)], \mathcal{H}_{\iota,1}$ is bounded and satisfies $|\mathcal{H}_{\iota,1}| \leq \bar{\mathcal{H}}_{\iota,1}$ for $\bar{\mathcal{H}}_{\iota,1} > 0$.

Since $\aleph_{\iota,1}(t) \in (-1, 1)$, it follows that $(1/(1 - \aleph_{\iota,1}^2)) \geq 1$, then one has

$$\begin{aligned}\dot{\mathcal{E}}_{\iota,1}(\mathfrak{X}_{\iota,1}) &\geq \lambda_{\iota,1}\frac{\aleph_{\iota,1}^2(t)}{1-\aleph_{\iota,1}^2(t)}\Psi_{\iota,1}^{-1}(b_\iota + p_\iota) - \frac{\varphi_{\iota,1}}{2}\aleph_{\iota,1}^2\\ &\quad - \frac{1}{2\varphi_{\iota,1}}\bar{\mathcal{H}}_{\iota,1}^2 \end{aligned}\tag{62}$$

$$\begin{aligned}&\geq \left(\lambda_{\iota,1}\underline{\Psi}_{\iota,1}^{-1}(b_\iota + p_\iota) - \frac{\varphi_{\iota,1}}{2}\right)\aleph_{\iota,1}^2(t) - \frac{1}{2\varphi_{\iota,1}}\bar{\mathcal{H}}_{\iota,1}^2,\\ &\geq -(-\mathfrak{y}_{\iota,1}\aleph_{\iota,1}^2(t) + \mathfrak{d}_{\iota,1})\end{aligned}\tag{63}$$

where $\mathfrak{y}_{\iota,1} = \lambda_{\iota,1}\underline{\Psi}_{\iota,1}^{-1}(b_\iota + p_\iota) - (\varphi_{\iota,1}/2) > 0$, and $\mathfrak{d}_{\iota,1} = (1/2\varphi_{\iota,1})\bar{\mathcal{H}}_{\iota,1}^2$.

By appropriately selecting $\varphi_{\iota,1}$ and $\lambda_{\iota,1}$ such that $\mathfrak{y}_{\iota,1} > \mathfrak{d}_{\iota,1}$, the time derivative of $\mathcal{E}_{\iota,1}(\mathfrak{X}_{\iota,1})$ satisfies

$$\begin{aligned}\dot{\mathcal{E}}_{\iota,1}(\mathfrak{X}_{\iota,1}) &\geq -\mathfrak{y}_{\iota,1}(1 - \aleph_{\iota,1}^2(t)) + \mathfrak{D}_{\iota,1}\\ &\geq -\varpi_{\iota,1}(\mathcal{E}_{\iota,1}(\mathfrak{X}_{\iota,1}))\end{aligned}\tag{64}$$

where $\mathfrak{D}_{\iota,1} = \mathfrak{y}_{\iota,1} - \mathfrak{d}_{\iota,1}$, and $\varpi_{\iota,1}(\cdot) \in \mathcal{K}_\infty^e$.

Define

$$\mathcal{E}_1(\mathfrak{X}_1) = \sum_{\iota=1}^{N}\mathcal{E}_{\iota,1}.$$

From (64), it follows directly that

$$\dot{\mathcal{E}}_1(\mathfrak{X}_1) \geq -\varpi_1(\mathcal{E}_1(\mathfrak{X}_1)) \tag{65}$$

where $\varpi_1(\cdot) = \max\{\varpi_{1,1}, \ldots, \varpi_{N,1}\}$.

Step $\ell(\ell = 2, \ldots, n-1)$: Define a continuously differentiable function

$$\mathcal{E}_{\iota,\ell}(\mathfrak{X}_{\iota,\ell}) = \frac{1}{2}\left(1 - \aleph_{\iota,\ell}^2(t)\right).$$

Taking the time derivative of $\mathcal{E}_{\iota,\ell}(\mathfrak{X}_{\iota,\ell})$ yields

$$\begin{aligned}\dot{\mathcal{E}}_{\iota,\ell}(\mathfrak{X}_{\iota,\ell}) &= -\aleph_{\iota,\ell}(t)\dot{\aleph}_{\iota,\ell}(t),\\ &= -\aleph_{\iota,\ell}(t)[-\lambda_{\iota,\ell}\frac{\aleph_{\iota,\ell}(t)}{1-\aleph_{\iota,\ell}^2(t)}\Psi_{\iota,\ell}^{-1} + \mathcal{H}_{\iota,\ell}(t)]\end{aligned}$$

where $\mathcal{H}_{\iota,\ell}(t) = \Psi_{\iota,\ell}^{-1}(t)[(\zeta_{\iota,\ell+1}(t) - (1 - \aleph_{\iota,\ell}^2(t)/2\Psi_{\iota,\ell}^{-1}(t))\hat{\mathcal{Q}}_{a,\iota\ell}^T\mathfrak{W}_{\iota,\ell}(\aleph_{\iota,\ell}) + \varrho_{\iota,\ell}(t)) - \dot{\Psi}_{\iota,\ell}\aleph_{\iota,\ell}(t)], \mathcal{H}_{\iota,\ell}$ is bounded and satisfies $|\mathcal{H}_{\iota,\ell}| \leq \bar{\mathcal{H}}_{\iota,\ell}$ for $\bar{\mathcal{H}}_{\iota,\ell} > 0$.

Similar to (62), one can obtain

$$\begin{aligned}\dot{\mathcal{E}}_{\iota,\ell}(\mathfrak{X}_{\iota,\ell}) &\geq \lambda_{\iota,\ell}\frac{\aleph_{\iota,\ell}^2(t)}{1-\aleph_{\iota,\ell}^2(t)}\Psi_{\iota,\ell}^{-1} - \frac{\varphi_{\iota,\ell}}{2}\aleph_{\iota,\ell}^2 - \frac{1}{2\varphi_{\iota,\ell}}\bar{\mathcal{H}}_{\iota,\ell}^2,\\ &\geq \left(\lambda_{\iota,\ell}\underline{\Psi}_{\iota,\ell}^{-1} - \frac{\varphi_{\iota,\ell}}{2}\right)\aleph_{\iota,\ell}^2(t) - \frac{1}{2\varphi_{\iota,\ell}}\bar{\mathcal{H}}_{\iota,\ell}^2,\\ &\geq -(-\mathfrak{y}_{\iota,\ell}\aleph_{\iota,\ell}^2(t) + \mathfrak{d}_{\iota,\ell})\end{aligned}\tag{66}$$

where $\mathfrak{y}_{\iota,\ell} = \lambda_{\iota,\ell}\underline{\Psi}_{\iota,\ell}^{-1} - (\varphi_{\iota,\ell}/2) > 0$, and $\mathfrak{d}_{\iota,\ell} = (1/2\varphi_{\iota,\ell})\bar{\mathcal{H}}_{\iota,\ell}^2$.

With suitable choices of $\varphi_{\iota,\ell}$ and $\lambda_{\iota,\ell}$ ensuring $\mathfrak{y}_{\iota,\ell} > \mathfrak{d}_{\iota,\ell}$, the derivative of $\mathcal{E}_{\iota,\ell}(\mathfrak{X}_{\iota,\ell})$ satisfies

$$\begin{aligned}\dot{\mathcal{E}}_{\iota,\ell}(\mathfrak{X}_{\iota,\ell}) &\geq -\mathfrak{y}_{\iota,\ell}(1 - \aleph_{\iota,\ell}^2(t)) + \mathfrak{D}_{\iota,\ell},\\ &\geq -\varpi_{\iota,\ell}(\mathcal{E}_{\iota,\ell}(\mathfrak{X}_{\iota,\ell}))\end{aligned}\tag{67}$$

where $\mathfrak{D}_{\iota,\ell} = \mathfrak{y}_{\iota,\ell} - \mathfrak{d}_{\iota,\ell}$, and $\varpi_{\iota,\ell}(\cdot) \in \mathcal{K}_\infty^e$.

Let

$$\mathcal{E}_\ell(\mathfrak{X}_\ell) = \sum_{\iota=1}^{N}\mathcal{E}_{\iota,\ell}.$$

According to (67), the time derivative of $\mathcal{E}_\ell(\mathfrak{X}_\ell)$ satisfies

$$\dot{\mathcal{E}}_\ell(\mathfrak{X}_\ell) \geq -\varpi_\ell(\mathcal{E}_\ell(\mathfrak{X}_\ell)) \tag{68}$$

where $\varpi_\ell(\cdot) = max\{\varpi_{1,\ell}, \ldots, \varpi_{N,\ell}\}$.

Step n: Consider a continuously differentiable function

$$\mathcal{E}_{\iota,n}(\mathfrak{X}_{\iota,n}) = \frac{1}{2}\left(1 - \aleph_{\iota,n}^2(t)\right).$$

Taking the time derivative of $\mathcal{E}_{\iota,n}(\mathfrak{X}_{\iota,n})$ yields

$$\begin{aligned}\dot{\mathcal{E}}_{\iota,n}(\mathfrak{X}_{\iota,n}) &= -\aleph_{\iota,n}(t)\dot{\aleph}_{\iota,n}(t),\\ &= -\aleph_{\iota,n}(t)[-\lambda_{\iota,n}\frac{\aleph_{\iota,n}(t)}{1-\aleph_{\iota,n}^2(t)}\Psi_{\iota,n}^{-1} + \mathcal{H}_{\iota,n}(t)]\end{aligned}$$

where $\mathcal{H}_{\iota,n}(t) = \Psi_{\iota,n}^{-1}(t)(b_\iota\tilde{\mathfrak{U}}_{\iota,f} - (1 - \aleph_{\iota,n}^2(t)/2\Psi_{\iota,n}^{-1}(t))\hat{\mathcal{Q}}_{a,\iota n}^T\mathfrak{W}_{\iota,n}(\aleph_{\iota,n}) + \varrho_{\iota,n}(t) - \dot{\Psi}_{\iota,n}\aleph_{\iota,n}(t)), \mathcal{H}_{\iota,n}$ is bounded and satisfies $|\mathcal{H}_{\iota,n}| \leq \bar{\mathcal{H}}_{\iota,n}$ for $\bar{\mathcal{H}}_{\iota,n} > 0$.

Similar to (62) and (66), one can obtain

$$\dot{\mathcal{E}}_{\iota,n}(\mathfrak{X}_{\iota,n}) \geq \lambda_{\iota,n}\frac{\aleph_{\iota,n}^2(t)}{1-\aleph_{\iota,n}^2(t)}\Psi_{\iota,n}^{-1} - \frac{\varphi_{\iota,n}}{2}\aleph_{\iota,n}^2 - \frac{1}{2\varphi_{\iota,n}}\bar{\mathcal{H}}_{\iota,n}^2$$

$$\geq \left(\lambda_{\iota,n}\underline{\Psi}_{\iota,n}^{-1} - \frac{\varphi_{\iota,n}}{2}\right)\aleph_{\iota,n}^2(t) - \frac{1}{2\varphi_{\iota,n}}\bar{\mathcal{H}}_{\iota,n}^2$$
$$\geq -(-\mathfrak{y}_{\iota,n}\aleph_{\iota,n}^2(t) + \mathfrak{d}_{\iota,n})$$

where $\mathfrak{y}_{\iota,n} = \lambda_{\iota,n}\underline{\Psi}_{\iota,n}^{-1} - (\varphi_{\iota,n}/2) > 0$, and $\mathfrak{d}_{\iota,n} = (1/2\varphi_{\iota,n})\bar{\mathcal{H}}_{\iota,n}^2$.

With suitable choices of $\varphi_{\iota,n}$ and $\lambda_{\iota,n}$ ensuring $\mathfrak{y}_{\iota,n} > \mathfrak{d}_{\iota,n}$, one can get

$$\begin{aligned}\dot{\mathcal{E}}_{\iota,n}(\mathfrak{X}_{\iota,n}) &\geq -\mathfrak{y}_{\iota,n}(1 - \aleph_{\iota,n}^2(t)) + \mathfrak{D}_{\iota,n},\\ &\geq -\varpi_{\iota,n}(\mathcal{E}_{\iota,n}(\mathfrak{X}_{\iota,n}))\end{aligned} \tag{69}$$

where $\mathfrak{D}_{\iota,n} = \mathfrak{y}_{\iota,n} - \mathfrak{d}_{\iota,n}$, and $\varpi_{\iota,n}(\cdot) \in \mathcal{K}_{\infty}^{e}$.

Let

$$\mathcal{E}_n(\mathfrak{X}_n) = \sum_{\iota=1}^{N} \mathcal{E}_{\iota,n}.$$

According to (69), the time derivative of $\mathcal{E}_n(\mathfrak{X}_n)$ satisfies

$$\dot{\mathcal{E}}_n(\mathfrak{X}_n) \geq -\varpi_n(\mathcal{E}_n(\mathfrak{X}_n)) \tag{70}$$

where $\varpi_n(\cdot) = \max\{\varpi_{1,n}, \ldots, \varpi_{N,n}\}$.

Combining (60), (65), (68), and (70), one obtains

$$\dot{\mathcal{E}}(\mathfrak{X}_1, \ldots, \mathfrak{X}_n) \geq -\varpi\left(\mathcal{E}(\mathfrak{X}_1, \ldots, \mathfrak{X}_n)\right) \tag{71}$$

where $\varpi(\mathcal{E}) = \max\{\varpi_1(\mathcal{E}_1), \ldots, \varpi_n(\mathcal{E}_n)\}$.

Based on Definition (1), Lemma (1), and (71), the function $\mathcal{E}(\mathfrak{X}_1, \ldots, \mathfrak{X}_n)$ qualifies as a CBF for the MASs (4), thereby ensuring that the set $\mathcal{G}$ is forward invariant. ■

Remark 4: To avoid overly aggressive control actions and numerical sensitivity near the safety boundary, we adopt a smooth quadratic CBF $\mathcal{E} = \sum_{\iota=1}^{N}\sum_{\ell=1}^{n}(1/2)\left(1 - \aleph_{\iota,\ell}^2\right)$, whose derivative $(\partial\mathcal{E}_{\iota,\ell}/\partial\aleph_{\iota,\ell}) = -\aleph_{\iota,\ell}$ remains bounded even as $|\aleph_{\iota,\ell}| \to 1$. Thus, safety is guaranteed by enforcing the CBC (forward invariance) rather than relying on a steep boundary gradient.

Remark 5: While standard funnel control [45], [46] can ensure forward invariance of the error bounds, it often lacks a formal safety certification through the CBF framework. This work explicitly establishes the connection between CBF and funnel functions, providing a concrete CBF construction that rigorously guarantees forward invariance and system safety, even in the presence of faults.

Remark 6: In contrast to traditional QP-based methods for ensuring safety [41], [42] whose computational complexity scales with the number of agents and requires solving an optimization problem online, the proposed method utilizes a CBF framework with fixed complexity throughout the backstepping procedure. This yields a low-complexity design that is suitable for real-time implementation and enhances scalability in large-scale MASs.

V. Simulation

To validate the proposed control strategy, a simulation is conducted on a MAS composed of robotic arms, whose dynamics can be modeled as in [47]

$$\mathfrak{M}(t)\ddot{\vartheta}(t) = -\frac{1}{2}\mathfrak{m}\mathfrak{g}\mathfrak{L}\sin(\vartheta) - \mathfrak{P}(\dot{\vartheta}) + \mathfrak{U}(t) + \varrho(t). \tag{72}$$

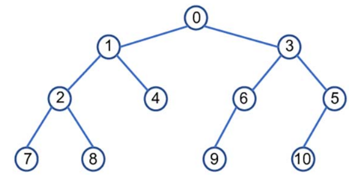
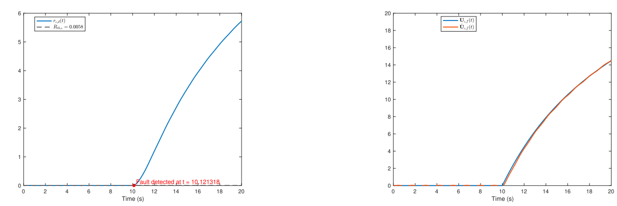
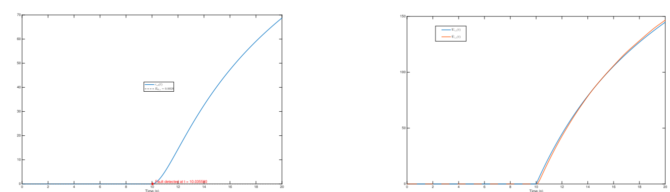


Fig. 2. Communication topology.



(a) (b)

Fig. 3. (a) Residual signal in Scenario 1. (b) FE trajectories in Scenario 1.



(a) (b)

Fig. 4. (a) Residual signal in Scenario 2. (b) FE trajectories in Scenario 2.

Here, $\vartheta(t)$ represents the joint angle, $\mathfrak{M}(t)$ is the moment of inertia, $\mathfrak{m}$ and $\mathfrak{L}$ denote the mass and length of the manipulator, and $\mathfrak{g}$ is the acceleration due to gravity. Additionally, $\mathfrak{U}(t)$ is the control torque, and $\varrho(t)$ accounts for any external disturbances. The water torque $\mathfrak{P}$ is expressed as $\mathfrak{P}(\dot{\vartheta}) = \bar{\mathfrak{C}}_1\dot{\vartheta}(t) + \bar{\mathfrak{C}}_2\dot{\vartheta}(t)$, where $\bar{\mathfrak{C}}_1$ and $\bar{\mathfrak{C}}_2$ are constants.

By setting the states as $\mathfrak{X}_1(t) = \vartheta(t)$ and $\mathfrak{X}_2(t) = \dot{\vartheta}(t)$, the dynamics for agent i are expressed as follows:

$$\begin{aligned}\dot{\mathfrak{X}}_{\iota,1}(t) &= \mathfrak{X}_{\iota,2}(t) + \wp_{\iota,1}(t) + \varrho_{\iota,1}(t)\\ \dot{\mathfrak{X}}_{\iota,2}(t) &= b_\iota\mathfrak{U}_\iota(t) + \wp_{\iota,2}(t) + \varrho_{\iota,2}(t)\\ y_\iota(t) &= \mathfrak{X}_{\iota,1}(t)\end{aligned}$$

where $\wp_{\iota,1}(t) = 0$, $\wp_{\iota,2}(t) = -(\mathfrak{m}\mathfrak{g}\mathfrak{L}/2\mathfrak{M})\sin(\mathfrak{X}_{\iota,1}(t)) - (\mathcal{C}/\mathfrak{M})\mathfrak{X}_{\iota,2}(t)$, b_ι is defined as $(1/\mathfrak{M})$, and the disturbances $\varrho_{\iota,1}(t)$ and $\varrho_{\iota,2}(t)$ are both modeled as $\varrho_{\iota,1}(t) = \varrho_{\iota,2}(t) = 0.04\cos(3t)$. The topology between the agents and the leader in the MASs is illustrated in Fig. 2.

Simulation parameters are set as $\mathfrak{M} = 2$, and $(\mathfrak{m}\mathfrak{g}\mathfrak{L}/2\mathfrak{M}) = (\mathcal{C}/\mathfrak{M}) = 0.2$. The leader's output is defined as $y_0(t) = \sin(0.5t)$. To assess the AFTC performance under different fault severities, an actuator fault is injected at $t = 10$ s with two test scenarios. In Scenario 1, the fault is modeled as $\mathfrak{U}_f(t) = 5[0.1(t-10) + 0.2\sin(0.1(t-10)) - 2e^{-0.2(t-10)} + 2]$. In Scenario 2, a larger-magnitude fault is considered: $\mathfrak{U}_f(t) = 50[0.1(t-10) + 0.2\sin(0.1(t-10)) - 2e^{-0.2(t-10)} + 2]$, which provides a more stringent validation

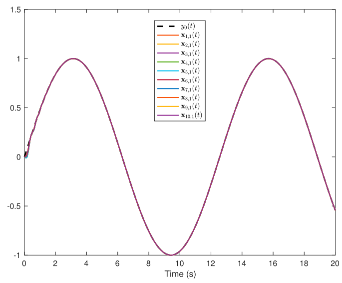


Fig. 5. Leader-follower tracking performance.

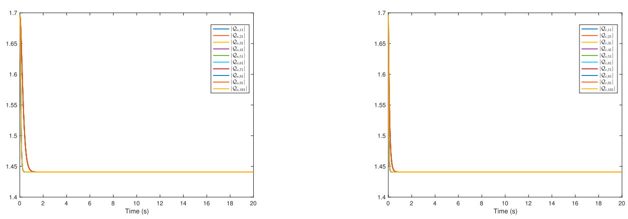


Fig. 6. Weight update laws of the actor-critic for first step.

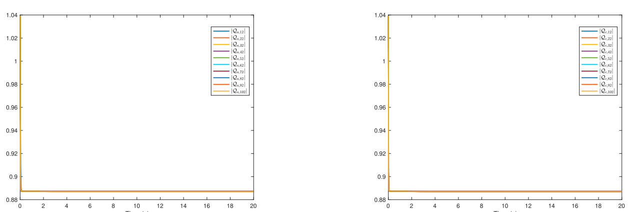


Fig. 7. Weight update laws of the actor-critic for second step.

of fault accommodation. The initial conditions are chosen as $\mathfrak{X}_{\iota,1}(0)=0$, $\mathfrak{X}_{\iota,2}(0)=0$, with $\lambda_\iota=1$, $\sigma_\iota=18$, $G_\iota=40$, and $L_\iota=30.201657$.

Figs. 3(a) and 4(a) show the residual signal $r_{\iota,d}(t)$ under two fault-magnitude scenarios. For both cases, the fault is successfully detected shortly after its injection at $t=10s$. In Scenario 1, the detection time is $t=10.121318s$, while in Scenario 2, the residual rises more rapidly and crosses the threshold earlier, indicating improved detectability for more severe faults. Figs. 3(b) and 4(b) further depict the actual fault $\mathfrak{U}_{\iota,f}(t)$ and its estimate $\hat{\mathfrak{U}}_{\iota,f}(t)$ for both scenarios, where the estimator closely tracks the fault profile and preserves high estimation accuracy despite the increased fault amplitude. Fig. 5 displays the leader-follower tracking performance, showing that all agents successfully track the leader's trajectory despite the fault. Figs. 6 and 7 depict the convergence of the actor and critic network weights.

The funnel functions are set to $\Psi_{\iota,1}(t)=(0.55-0.02)e^{-t}+0.02$ and $\Psi_{\iota,2}(t)=(1-0.05)e^{-1.5t}+0.05$, with the class $\mathcal{K}_\infty^e$ function $\varpi(s)=5s$. Fig. 8 confirms that the consensus errors remain strictly within their respective funnel boundaries, satisfying the prescribed performance. Fig. 9 shows the evolution of the CBFs $\mathcal{E}_{\iota,1}$ and $\mathcal{E}_{\iota,2}$, which remain positive. Fig. 10(a) verifies that the CBC is satisfied for all time, confirming that $\mathcal{E}$ is a CBF and that the safe set $\mathcal{G}$ is forward invariant.

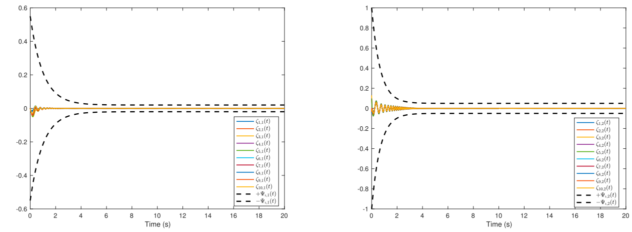


Fig. 8. Trajectories of the errors $\zeta_{\iota,1}(t)$, $\zeta_{\iota,2}(t)$ and the funnel boundaries $\Psi_{\iota,1}(t)$, $\Psi_{\iota,2}(t)$.

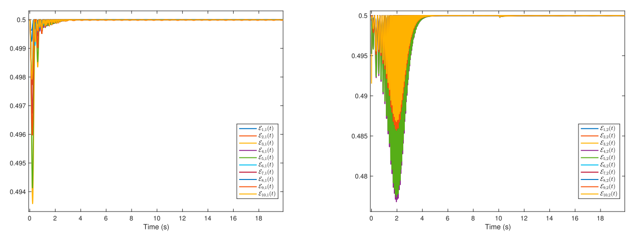
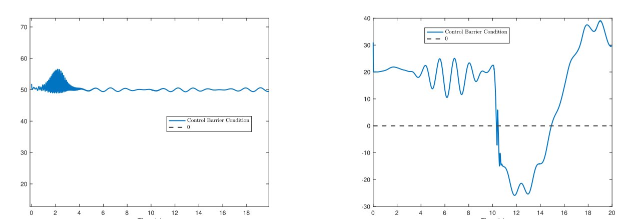


Fig. 9. CBF $\mathcal{E}_{\iota,1}$ and $\mathcal{E}_{\iota,2}$.



(a) (b)

Fig. 10. CBC comparisons: (a) CBC under the proposed design; and (b) CBC under the method in [45].

To highlight the safety contribution, a comparison is made with the method from [45], which uses funnel control without an explicit safety analysis. Fig. 10(b) shows that for the controller in [45], the CBC is violated during certain time intervals, demonstrating its lack of a formal safety guarantee.

VI. Conclusion

This article has presented a novel framework for the optimal and safe tracking control of MASs subject to actuator faults. By integrating RL, AFTC, and CBFs, the proposed strategy achieves robust performance and guarantees safety. The AFTC module ensures rapid fault detection and compensation, while the RL-based actor-critic structure learns the optimal control policy online. Crucially, the CBF derived from funnel functions provides a rigorous safety guarantee without resorting to computationally expensive online optimization. This low-complexity design ensures scalability and real-time applicability in large-scale systems. Simulation results have validated the effectiveness of the proposed approach, demonstrating its ability to maintain stable consensus tracking and ensure system safety in the presence of significant faults and disturbances. Future work may explore the extension of this framework to address more complex system dynamics and communication constraints. Future work will focus on extending the proposed framework to more general agent dynamics and to communication-impaired networks with delays or cyberattacks. In addition, robustness to sensor degradation and experimental validation on multirobot platforms will be investigated.

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